# SMEs and Public Procurement: the Costs of Restricting Tenders*

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**Abstract**

This paper estimates the costs of restricting public tenders to small and medium-sized enterprises (SMEs). I exploit a quasi-experimental policy change in Sao Paulo, Brazil: until March 2018, medical supplies (group 65) were exempt from mandatory SME-only tenders, while all other product groups faced the restriction. Using a difference-in-differences in reverse (DiDiR) design, I find that open tenders yielded negotiated prices 12–13% lower, attracted approximately 19% more participating firms, and generated approximately 16% more valid bids than SME-restricted tenders. The fiscal cost of the restriction for group 65 alone amounts to R$84.5–85.8 million in nominal terms (approximately US$17 million) over the 18-month pre-period, or about 12% of procurement value for this product group. Winning firms under open tenders were located 5–11 km further from buyers, suggesting that SME restrictions do encourage local sourcing. Results are robust to alternative clustering, winsorization, Lee bounds, and sensitivity analysis.

**Keywords:** public procurement, SMEs, policy costs, restricted public tenders.

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# 1. Introduction

Small and medium-sized enterprises are economically important in countries worldwide. In Europe, for instance, SMEs represent nearly 99.8% of all registered companies, accounting for more than half of the European GDP and two-thirds of all jobs in the private sector (PwC, 2014). Similar figures can be observed in Brazil, where approximately 97% of all firms are SMEs, representing approximately 50% of Brazil's formal jobs (Bastos et al., 2018).

In recent years, governments worldwide have implemented public policies that favor SMEs based on the potential or actual benefits that these policies can bring to the economy.

The literature widely acknowledges that SMEs have massive potential for job creation, local development, and innovation (SIGMA, 2016). However, there is limited evidence on the social costs of such policies, since most research and case studies do not address this point. This paper exploits a quasi-experimental variation from a public SME-related program to estimate the costs of incentivizing restricted public tenders to SMEs in Sao Paulo, Brazil.

The use of public procurement as a policy tool has been a major trend worldwide (Thai, 2017; Bosio et al., 2022). Policies that favor sustainable or 'green' procurement or utilize social criteria to restrict bids to a target group of sellers are prime examples of public procurement policy that aims to achieve social and economic outcomes.

One of the most widespread practices to promote local development through public procurement is the restriction of public tenders to SMEs. The significant presence of SMEs in the economy suggests that these companies are an essential channel through which to infuse economic development (Mel et al., 2008; Freedman, 2013), although there is evidence that the effectiveness of this mechanism depends on the quality of management practices (McKenzie and Woodruff, 2015) or how capital is provided through investment in SMEs (Fafchamps et al., 2014).

The primary justifications for implementing policies that favor SMEs in public procurement are related to the various entry barriers to public tender that SMEs face (OECD, 2018; Loader, 2015; Hoekman and Tas, 2020). SMEs typically lack the administrative capacity to navigate complex bidding procedures, face higher per-unit compliance costs, and may be unable to meet minimum lot-size requirements designed for larger suppliers. Better access to the public procurement market might catalyze SMEs' productive potential, especially in contexts with demand constraints (Cardoza et al., 2016; Ferraz et al., 2016; Szerman, 2023). It may expand government networks of goods and services providers, thus enhancing the competition among firms and enabling public acquisitions at more competitive prices (Loader, 2013).

Despite the potential positive effects generated by favoring SMEs in public tenders, such a policy may undermine the quality and efficiency of the public procurement process (Nakabayashi, 2013; Athey et al., 2013). Set-asides and bid preferences may substantially reduce procurement efficiency and revenue compared with open competition (Krasnokutskaya and Seim, 2011; Marion, 2007). First, restricting tenders to SMEs may lead to smaller-scale purchases per bid because SMEs generally cannot handle large orders.

Second, restricting public bid participation may harm firms' screening process and increase the likelihood of selecting fewer, less efficient firms or no firms at all (Nakabayashi, 2013; Loader, 2013; Szucs, 2024). When the pool of eligible bidders shrinks, the government loses its ability to leverage competitive pressure to drive down prices, and the remaining bidders may have less incentive to submit aggressive bids. Thus, either negotiated prices may be higher than usual or there may be a waste of public resources since planning and executing a bid is costly. However, there is evidence that the costs of negotiating and executing public contracts may be mitigated by public managers' contract management capabilities and private sellers' contract execution capabilities (Cabral, 2017).

Additionally, depending on the characteristics of their sector, SMEs may not reach competitive price-cost levels when providing a good or service to the government (Nakabayashi, 2013; Decarolis et al., 2024). Thus, organizing an unsuccessful tender means wasting resources; the government may incur severe costs with few results for SMEs or the local economy.

While there are numerous examples worldwide of policies favoring SMEs in public tenders, there is little direct evidence of the impacts of such policies on firms performance. Additionally, the costs of implementing such a policy are practically ignored in the literature. This gap in the evidence base is consequential: without credible estimates of the costs that SME set-asides impose on public procurement, policymakers cannot adequately weigh these costs against the distributional or developmental benefits that such policies are intended to deliver. Recent methodological advances in causal inference—including machine learning approaches for heterogeneity analysis (Athey et al., 2019) and sensitivity tools for assessing the robustness of parallel trends assumptions (Rambachan and Roth, 2023)—offer new opportunities to rigorously evaluate these policies, yet their application to procurement set-aside programs remains nascent.

This paper provides new evidence on this question by exploiting a quasi-experimental variation from a public SME-related program to estimate the costs of incentivizing the restriction of public tenders to SMEs in Sao Paulo, Brazil. I exploit the timing of a change in a policy of restricted SME tenders (March 2018) that affect only a specific group of items (group 65). The identification strategy simultaneously uses time and cross-sectional variations to estimate the effects of this policy shift.

The institutional setting provides unusually clean identification. Between August 2014 and February 2018, the state of Sao Paulo exempted group 65 (medical and hospital supplies) from the mandatory SME-only tender requirement that applied to all other product groups, based on a joint agreement between the state government and the audit court that health items constituted strategic goods warranting open competition. In March 2018, a legal opinion from a different oversight agency reversed this interpretation, subjecting group 65 to the same SME-favoring rules as all other groups. This policy change was driven by a legalistic reinterpretation of isonomy principles rather than by changes in procurement outcomes for group 65, making it plausibly exogenous to the outcomes studied.

However, examining how this institutional change occurred allows the costs of the SME policy to be assessed only indirectly. Instead of using a standard difference-in-differences (DiD) method, I use a variation of this method known as difference-in-differences in reverse (DiDiR), or 'time-reversed DiD' (Kim and Lee, 2019). In the DiD method, there is a control group that is never treated and a treatment group that is treated at some point in time. In the DiDiR method I employ, the control group is always treated (instead of always untreated), and the other group is the 'switched group,' subject to the change in policy.

In Sao Paulo, the government can execute public tenders for SMEs only. Between August 2014 and February 2018, the government's default choice consisted of executing SME-only tenders for all items with a value less than or equal to R$80,000, except for a group of items identified as code 65, which was only allowed to execute open tenders in this period.

The government can avoid restricted tenders for SMEs if it considers that any item in a bidding process falls within the exceptions provided for by law through a costly process of justification to its watchdogs. In case of any irregularity or failure to comply strictly with the law, public officers can be punished. From March 2018 on, after a change in the law's interpretation, group 65 became subject to the general rule just as any other group of purchased items. Thus, the 'always treated' group here consists of all groups of items but group 65, comprising the switched group.

Notably, DiDiR identifies pre-switch-period effects; i.e., it estimates effects for the past (Kim and Lee, 2019), assessing the costs of the policy switch indirectly. I estimate this pre-switch-period effect on the switched group, comparing the observed outcomes for group 65 before the shift in policy and the outcomes that would have occurred for this group if there had been opt-out costs before March 2018.

The DiDiR method reveals that the negotiated prices are, on average, between 12% and 13% lower for group 65 than other groups before March 2018 (approximately 13 log points across specifications). Moreover, in the pre-period, the number of participants in group 65 is approximately 19–20% higher than that in other groups of items in the short-term window. The number of valid bids follows the same pattern: there are approximately 16–17% more valid bids in group 65 than in the other groups of items. Finally, before the policy change, sellers that won tenders for items in group 65 were approximately 5–11 km further away from public buyer units (PBUs), depending on the specification. These price effects are concentrated among high-value items, where the pool of capable non-SME suppliers is broader and the efficiency gains from open competition are larger. These are nominal estimates; IPCA-deflated coefficients range from 7% to 12% (see Section 4.4). Translating the nominal estimates into monetary terms, the fiscal cost of the SME-only restriction for group 65 alone amounts to R$84.5–85.8 million (approximately US$17 million) over the 18-month pre-period—about 12% of this product group's total procurement value—underscoring the substantial efficiency losses that set-aside policies can impose on public procurement.

The results are supported by a comprehensive set of robustness checks. Placebo tests using fake treatment dates confirm that the estimated effects are not driven by pre-existing differential trends. Alternative clustering strategies, winsorization of outcomes, and randomization inference all corroborate the main findings. Extensions to IPCA-deflated real prices, procurement efficiency ratios, and winner composition analysis provide evidence on the channels through which the policy operates. Advanced econometric methods in the appendix provide further corroboration: Lee bounds demonstrate that the price effect survives correction for sample selection, quantile DiD reveals that the effect is concentrated at lower quantiles of the price distribution, and a causal forest analysis identifies item quantity as the key moderator of treatment effect heterogeneity.

This paper contributes to two strands of the literature. First, it adds to the growing body of evidence on the costs of restricting competition in public procurement (Bosio et al., 2022; Szucs, 2024; Decarolis et al., 2024; Bandiera et al., 2009; Best et al., 2023), providing estimates from a large developing economy where SME policies are especially prevalent. Second, it contributes to the literature on set-asides and bid preferences (Athey et al., 2013; Krasnokutskaya and Seim, 2011; Marion, 2007) by documenting how these costs vary across item types, offering a richer picture of how procurement efficiency losses distribute across heterogeneous goods. A distinguishing feature of this paper is its methodological breadth: beyond the standard reduced-form analysis, I deploy a suite of advanced methods—including parallel trends sensitivity analysis, sample selection bounds, causal forests, quantile regressions, and mechanism decomposition—to stress-test the findings from multiple angles.

The remainder of this paper is organized as follows. Section 2 provides a brief overview of the institutional background related to SME law and public procurement in Sao Paulo, Brazil. Section 3 describes the datasets and sample definitions. Section 4 presents the empirical analysis, including heterogeneity analysis, robustness checks, and a fiscal cost quantification. Section 5 concludes with policy implications.

# 2. Institutional Background

This section provides a brief institutional background on SMEs' participation in the context of public tenders in Sao Paulo, Brazil. I focus on the details that are most pertinent for the empirical analysis.

First, I briefly discuss general aspects of public procurement and the legislation on micro and small businesses in Brazil, highlighting the established criteria for the occurrence of SME-only public tender. Finally, I describe how the state of Sao Paulo applied this law in accordance with its own understanding of exclusive tenders for health-items.

## 2.1. Public Procurement and SME Law in Brazil

Public procurement constitutes a relevant part of economies worldwide (Bosio et al., 2022). In 2016, OECD countries spent an average of approximately 12% of GDP on public procurement, while in Brazil, this proportion was approximately 10% in the same year. Recent evidence from Brazil shows that procurement institutions and their enforcement have significant effects on firm dynamics and employment (Szerman, 2023; Colonnelli and Prem, 2022).

As in many other countries, Brazilian law establishes as a general rule that all purchases, services, and works hired by the public administration should be subject to a public tender. Federal Law 8,666/1993 institutes a general framework applicable to all public bids in the country, and all three government branches must adhere to this framework.

Entities directly or indirectly controlled by the federal, state, or municipal governments must comply with the government procurement rules. Federal, state and municipal governments, autonomous government entities, public foundations, regulatory agencies, state-owned companies, and mixed capital companies controlled by the government are subject to these rules. These entities are known as public buyer units (PBUs).

Although the public administration may decide to make purchases centrally, in Brazil, almost all acquisitions are decentralized and made by PBUs. A ministry or bureau may consist of many PBUs that have budgetary autonomy and make purchases from private companies. PBUs may contract a wide variety of products and services from private companies, including engineering and infrastructure work. However, this paper focuses on analyzing the acquisition of common and standardized goods.

The primary purpose of a bidding process conducted by a PBU is to seek the best contract possible for the government. The Brazilian public procurement law provides guidelines on how the procurement process should be organized and executed. In some cases, public tenders for SMEs are subject to different treatment.

The Brazilian federal SME law was created in 2006 to regulate favored, simplified and differentiated treatment for this sector, as provided for in the Federal Constitution. This law's explicit goal was to promote SMEs' economic and social development and competitiveness as a strategy for job creation, income distribution, social inclusion, reduced informality, and a strengthened economy.

The Brazilian SME law adopts the following classification for companies, according to their annual gross revenue: (i) microbusiness: annual gross revenue of R$360,000 or less (roughly US$72,000); and (ii) small business: annual gross revenue greater than R$360,000 and less than or equal to R$4,800,000 (between US$72,000 and US$960,000). In this paper, these companies are referred to as SMEs.

SMEs enjoy many benefits provided by law, including tax benefits and fewer bureaucratic requirements to adhere to. In addition, public tenders held at the federal, state, and municipal levels can grant differentiated and privileged treatment to SMEs to promote economic and social development, increase public policies' efficiency, and stimulate technological innovation.

The content of the SME law, in its 2006 version, indicated that the public administration *could* create tenders exclusively for the participation of SMEs in purchases in which the item value was up to R$80,000 (approximately US$16,000). Thus, choosing tenders for SMEs only was optional for PBUs.

However, the federal SME law underwent a significant change from SMEs' exclusivity in tenders in 2014. The term "could" has been replaced with "must," making it mandatory to execute exclusive public tenders for SMEs up to a value per item of R$80,000.

The law changed PBUs' default choice if the item value fell below the threshold of eighty thousand reais: previously, the default choice was open bids. As of 2014, the standard option for PBUs is to execute public tenders for SMEs only, unless the bid's conditions fall within the exceptions provided for in the updated legislation.

PBUs can avoid restricted bids if at least one of the following conditions is met: (i) there are two or fewer potential competing SME suppliers that are locally or regionally based and able to comply with the notice requirements, and (ii) PBUs consider that the differentiated and simplified treatment for SMEs might not be advantageous for the public administration. Thus, PBUs choose whether the public tender is restricted, but they must justify their choices to their watchdogs, such as audit courts or the judiciary.

On the one hand, this discretion provided for by law can be beneficial since PBUs can more efficiently choose the bidding type to be carried out. However, there are costs involved in the process of avoiding bids restricted to SMEs. For each bidding procedure, PBUs must create an extensive report listing in detail the reasons that justify the use of an open bidding process to the detriment of a bidding process restricted to SMEs.

Additionally, these PBU justifications are subject to scrutiny by both the audit courts and the judiciary. If these bodies consider the arguments unfounded or insufficient, administrative proceedings and punishments may be brought against the public agents responsible for planning and executing the bid in question. Thus, this discretion brings costs to PBUs. I call these costs associated with avoiding SME-only tenders *opt-out costs*.

## 2.2. SME-only Public Tenders: Group 65 As an Exception

Sao Paulo is the wealthiest and most populous state in Brazil. This state accounts for approximately 23% of the total population and nearly one-third of Brazil's GDP (nearly US$500 billion in 2018). This amount is equivalent to the GDP of countries such as Sweden, Poland, and Belgium and more than twice the GDP of Portugal, Greece, and Finland. The state of Sao Paulo has a diversified economy driven by the automobile, textile, chemical, aeronautical, and computer industries, in addition to services such as finance and agriculture.

Since 2005, all PBUs in the state of Sao Paulo have been required to purchase common goods and services through Bolsa Eletrônica de Compras (BEC), an electronic purchasing platform. BEC operates as a centralized electronic reverse auction system (*pregão eletrônico*), in which registered suppliers submit progressively lower bids in real time until the auction closes, facilitating transparent price competition. This design is comparable to other electronic procurement platforms adopted worldwide, such as Chile's ChileCompra and South Korea's KONEPS, which similarly aim to reduce transaction costs, increase supplier participation, and improve price outcomes through digitized and standardized processes. The BEC figures are revealing: in 2019, approximately R$13 billion (about US$3 billion) in trade was conducted on this e-platform. Since its implementation in 2005, BEC has moved more than R$105 billion (about US$20 billion) in negotiations; 860,000 purchase offers have been made, and approximately 4.8 million items have been sold.

Despite being subject to federal laws, Brazilian states have the prerogative to regulate or interpret these laws' specific elements. The state of Sao Paulo, for example, has a specific interpretation of how to apply SME law in public procurement.

Since BEC's implementation, the Sao Paulo state government has considered the group of items consisting of health-related products, including medication and hospital supplies (code 65), as a strategic set of items in the public procurement process. For example, bidding procedures related to medication can be carried out only with dynamic reverse auctions (*pregão*) or sealed bidding (*convite*). Direct negotiation (*dispensa de licitação*), which is very common in the purchase of other types of common goods, has always been prohibited for group 65 in Sao Paulo.

Between 2006 and 2014, when the first version of the SME law was enforced, PBUs located in the state of Sao Paulo had the default choice to hold open tenders; it was optional to set procedures restricted to SMEs. During this period, in accordance with this legal arrangement, there was low adherence to restricted bids; items in these bids accounted for 6 to 13% of the total number of items bid for in the state of Sao Paulo. However, for group 65, there were no bids exclusively for SMEs in this same period. The government had an internal orientation to hold open tenders for group 65, with the explicit agreement of the audit court of the state of Sao Paulo (TCE-SP).

After 2014, with the update of the federal law on SMEs, the default choice was to execute SME-only tenders if the item value was less than or equal to R$80,000. In the state of Sao Paulo, if PBUs consider that any item in a bidding process falls within the exceptions provided for by law, they must justify in detail the reasons for the non-execution of an exclusive bid for SMEs through a report sent to TCE-SP.

This justification offered by PBUs to avoid restricted tenders not only requires excessive work effort for PBUs but also is subject to the scrutiny of the TCE-SP and the judiciary. Public officers can face punishment if there are irregularities or failure to comply strictly with the law.

With the introduction of considerable opt-out costs for PBUs, this incentive scheme appears to be effective in encouraging the choice of offering exclusive bids to SMEs: adherence to this procedure has increased from approximately 13% to almost 70%, on average, in subsequent periods.

However, between August 2014 and February 2018, bids related to group 65 did not change. The Sao Paulo state government and TCE-SP, in a joint agreement, used a specific interpretation of the new version of the federal SME law to leave group 65 as an exception. The federal law provides that the requirement of restricted bids to SMEs does not apply when "it is not advantageous to the public administration or represents a loss to public resources." Thus, using this guideline, which allows for a high degree of discretion, the state of Sao Paulo and TCE-SP operated on the interpretation that because group 65 constitutes a set of strategic items to meet such an essential public policy, it should always be subject to open bids. Thus, in this period, there was no bidding restricted to SMEs involving group 65.

However, in November 2017, a different control agency in the state of Sao Paulo, PGE-SP, issued a document containing a legal opinion that changed this interpretation. This document reinforces that the principle of isonomy in law enforcement should prevail in public procurement. Then, items in group 65 should be considered for public procurement purposes in the same way as other groups.

Hence, as of March 2018, the opt-out costs also apply to group 65. Between March 2018 and December 2019, adherence to exclusive bids for SMEs for this group was, on average, 43%. This lower proportion of restricted bids for SMEs for group 65 compared with other groups may be due to the existence of more oligopolies in this group of items, which include, for example, medication. Thus, in many cases, it is not possible to find at least three potential suppliers that are SMEs.

Two features of this institutional change are particularly important for the empirical strategy. First, the trigger for the policy change was a legalistic reinterpretation of the isonomy principle by PGE-SP, not a response to deteriorating procurement outcomes for group 65. There is no evidence that the decision to extend SME-only tender rules to group 65 was motivated by price levels, competition patterns, or supplier complaints in this specific product group. Second, the timing of the change was driven by bureaucratic processes within PGE-SP and its communication to PBUs, making it plausibly exogenous to the procurement outcomes studied in this paper. These features provide the institutional basis for the identifying assumption that, absent the policy change, group 65 and other groups would have followed parallel outcome trajectories in the post-period.

# 3. Data

This section describes the data source and details the sample characteristics used in the empirical section.

I use administrative data on bidding-level public procurement tenders of common goods and services in the state of Sao Paulo, Brazil, from January 2016 to December 2019. All transactions took place on the electronic procurement platform called BEC, available for all PBUs across the state. The SEFAZ/SP (State Finance Secretariat) is responsible for the operational management and centralization of BEC's bidding data. An important advantage of this data source is that BEC records the universe of standardized goods procurement in the state, eliminating selection concerns that arise in settings where data coverage is partial or voluntary.

On BEC, 1,344 PBUs regularly make purchases. These entities include state-level bureaus from the executive, legislative, and judiciary branches in the state of Sao Paulo as well as other affiliated entities, such as municipalities located in the state of Sao Paulo and private organizations. PBUs purchased 82,569 different items (goods) totaling 832,984 successful transactions from 2016 to 2019.

BEC has a very detailed catalog of standardized goods and services organized in three levels of detail: group, class, and item. For instance, health items are classified as group 65 (medical, dental, and hospital equipment and supplies). The item coded as 110639 refers to the drug 'Furosemide 40 milligrams, coated tablets, units', belonging to class

6531 (Medicines prescribed with or without ANVISA notification/registration) and group 65.

Data are organized by purchase offer (PO), the electronic document issued by the PBU that identifies and quantifies the goods and services that will be purchased. A PO is defined by a 22-character code and may contain one or more items listed, but each item has its own purchase process. Thus, the purchase of an item is uniquely identified by the combination of the PO and the purchased item codes (POI).

There is a crucial variable for the empirical section, defined from the item group codes. It is a binary variable that assumes the value of 1 if the item belongs to group 65 and 0 otherwise. The items in group 65 constitute the 'switched group', i.e., the set of items affected by the purchasing policy change. Group 65 accounted for almost 27% of all purchases from 2016 to 2019. All other groups of items comprise the 'control group.'

There are 75 groups of items, excluding group 65. Between 2016 and 2019, the most significant groups were groups 89 (food products), 75 (office supplies), 86 (computer products), and 79 (cleaning materials), representing 37.5% of the total purchases in this period. The diversity of the always-treated group—spanning food, office supplies, technology, and cleaning materials—mitigates the concern that the comparison group is idiosyncratic or dominated by a single product category.

For each POI, there is information about item quantities, bid prices (winners and losers), the number of participant firms, the number of bids, whether the public tender was successful or not, the identification of the PBU, and firms' and PBUs' location, among other variables.

The empirical analysis focuses on four main outcome variables. First, the *negotiated price* (in logs) captures the final price at which the item was transacted; this variable is available only for completed items. Second, the *number of participant firms* (in logs) measures how many distinct firms submitted at least one bid, regardless of whether the item was ultimately purchased. Third, the *number of valid bids* (in logs) counts the total number of qualifying bids received per item. Fourth, *distance* measures the geographic distance in kilometers between the winning firm's location and the PBU. The regressions control for the log quantity ordered, the type of tender (sealed bid vs. reverse auction), and include item-level fixed effects that absorb all time-invariant heterogeneity across items—including differences in markets, pricing norms, and typical supplier pools.

To ensure that the estimates are not driven by compositional changes in the sample, the analysis restricts attention to items that appear in both the pre- and post-periods within each time window. The three time windows (6, 12, and 18 months around the March 2018 cutoff) provide a check on the stability of the estimates as the sample expands. The price and distance regressions are estimated on the subsample of completed items (where a winning bid exists), while the participation and bid regressions use all items, including those where no firm was selected. An important implication of conditioning on completion is that the price and distance estimates may be affected by sample selection if the treatment itself influences whether items are successfully transacted; the Lee bounds analysis in the appendix directly addresses this concern. Additionally, the breadth of item groups in the always-treated category—spanning food products, office supplies, computer equipment, cleaning materials, and dozens of other categories—provides substantial cross-sectional variation that strengthens the external validity of the estimates beyond any single product market.

Table 1 presents descriptive statistics for the key variables used in the empirical analysis, disaggregated by group (Group 65 vs. Others) and period (Pre vs. Post the policy shift in March 2018).

Table 1: Descriptive Statistics (18-month window)

| | Group 65, Pre | | Group 65, Post | | Others, Pre | | Others, Post | |
|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD | Mean | SD |
| Price (levels) | 1,402.09 | 20,563.54 | 1,507.92 | 25,872.31 | 1,465.85 | 50,385.93 | 1,291.37 | 68,042.55 |
| Log price | 1.72 | 2.84 | 1.91 | 2.81 | 2.85 | 2.12 | 2.78 | 2.07 |
| Number of firms | 3.17 | 2.52 | 3.02 | 2.40 | 4.66 | 3.27 | 4.69 | 3.28 |
| Log number of firms | 0.88 | 0.73 | 0.84 | 0.71 | 1.30 | 0.72 | 1.30 | 0.72 |
| Number of valid bids | 11.10 | 15.49 | 10.99 | 16.51 | 10.13 | 15.74 | 10.47 | 16.87 |
| Log number of valid bids | 1.64 | 1.24 | 1.61 | 1.23 | 1.69 | 1.06 | 1.70 | 1.08 |
| Distance (km) | 238.54 | 270.42 | 215.08 | 239.63 | 164.78 | 182.23 | 170.28 | 186.07 |
| N (completed / all) | 65,606 | 77,206 | 66,822 | 85,181 | 246,277 | 287,500 | 271,011 | 323,836 |

*Notes:* Price and distance statistics computed on completed items (status = 1). Firm and bid statistics computed on all items.

# 4. Empirical Strategy and Results

This section is organized into two parts. First, I describe the method I use to estimate the impacts of the SME public procurement policy shift on the selected outcomes. I perform a DiD analysis, but unlike a standard DiD, the control group is always treated instead of always untreated (Kim and Lee, 2019). Then, I discuss the identifying assumptions in the context of the DiDiR I employ. In the second part, I discuss the main results.

## 4.1. Main Specification and Identification Strategy

The identification strategy exploits the timing of a change in the policy of restricted SME tenders (March 2018) that affect only a specific group of items (group 65). Thus, it is possible to simultaneously use time and cross-sectional variations to estimate the potential effects of this policy shift.

In a standard DiD with treatment $d$ and outcome $y$, there exists a group of units ($q = 1$) with its treatment changing from $d = 0$ to $d = 1$ at some date, and there exists another group ($q = 0$) in which $d = 0$ always. In this paper, however, the framework is slightly different: the group "$q = 0$" always has $d = 1$ instead of $d = 0$, and the group "$q = 1$", as in DiD, undergoes a switch in the treatment.

This variation of a standard DiD is known as difference-in-differences in reverse (DiDiR), or "time-reversed DiD" (Kim and Lee, 2019). This same framework can be found in (Kotchen and Grant, 2011; Chemin and Wasmer, 2009; Autor et al., 2006; Shapiro and Gentzkow, 2008; Monstad et al., 2008). Importantly, because the design involves a single treatment date and a clean two-group comparison, the estimation is not subject to the staggered-adoption concerns that may bias two-way fixed effects estimators with heterogeneous treatment effects (de Chaisemartin and D'Haultfoeuille, 2020; Roth et al., 2023).

As described in Section 3, from August 2014 to February 2018, PBUs faced opt-out costs to avoid tendering for SMEs only, with the exception of group 65 (with mandatory open tenders). From March 2018 to December 2019, PBUs were subject to opt-out costs for all groups of items purchased. Thus, the always treated group here consists of all groups of items but group 65, which comprises the switched group. Table 2 summarizes the above description.

Table 2: Description of groups in DiDiR

| Groups | $t = 1$ (before March 2018) | $t = 2$ (after March 2018) |
|---|---|---|
| Group 65 (switched group) | Opt-out costs $= 0$ | Opt-out costs $> 0$ |
| Others (always treated group) | Opt-out costs $> 0$ | Opt-out costs $> 0$ |

DiDiR identifies pre-switch-period effects; i.e., it estimates effects for the past (Kim and Lee, 2019). I estimate this pre-switch-period effect on the switched group, comparing the observed outcomes for group 65 before the policy shift and the outcomes that would have occurred for this group if there had been opt-out costs before March 2018 ($t = 1$).

Using subindex $p$ to denote purchase offer, $i$ to denote items, $t$ to denote months, and $g$ to denote groups of analysis, I estimate the following DiDiR model:

$$y_{pigt} = \eta_i + \gamma \,\mathrm{Pre}_t + \beta \left( g65_{pgt} \times \mathrm{Pre}_t \right) + x\,\delta + \varepsilon_{pigt} \qquad (1)$$

where $y$ is an outcome, $\eta_i$ represents item fixed effects, and *Pre* is a dummy variable with value 1 if it is a month before March 2018 and 0 otherwise. The variable $g65$ is binary with a value of 1 if it belongs to group 65 and 0 otherwise. The covariates are represented by $x$. The error $\varepsilon_{pigt}$ is clustered by item.

The coefficient of interest, $\beta$, captures the pre-switch-period effect of the shift in the SME tender policy on the outcomes for the switched group. Under the DiDiR framework, $\beta$ measures the difference in outcomes for group 65 relative to the counterfactual scenario in which group 65 had always faced the same opt-out costs as other groups. A negative $\beta$ for prices, for instance, indicates that prices were lower under open tenders than they would have been under SME-restricted tenders.

The central identifying assumption behind the empirical model is that, in the period after the shift in the tenders policy, the outcomes for group 65 and the set of all other groups of items would have followed a similar trajectory. This assumption is testable in the post-period (where both groups face the same policy regime) and is supported by the institutional argument that the policy change was driven by a legalistic reinterpretation rather than by changes in the outcomes themselves. Thus, it is necessary to check whether the outcome paths of the always treated and switched groups are parallel in the post-switch period.

The validity of this assumption of parallel trends in the post-switch period can be partially assessed by estimating the following nonparametric regression, similar to (Naritomi, 2019; Gallego et al., 2020):

$$y_{pgt} = \eta_g + \text{Semester}_t + \sum_{k=-3}^{3} \tau^t \Big( \beta \left( g65_{pgt} \times \text{Semester}_t \right) \Big) + \mu_{pgt} \tag{2}$$

where $\eta_g$ is the group fixed effects, and *Semester* is a set of dummies for each semester in this period. The error $\mu_{pgt}$ is clustered by the group of items. Figure 1 plots the coefficients (without a constant) and the 95 percent confidence intervals from estimating equation (2) with log prices as the dependent variable. The graphs for the other outcomes of interest are included in the appendix.

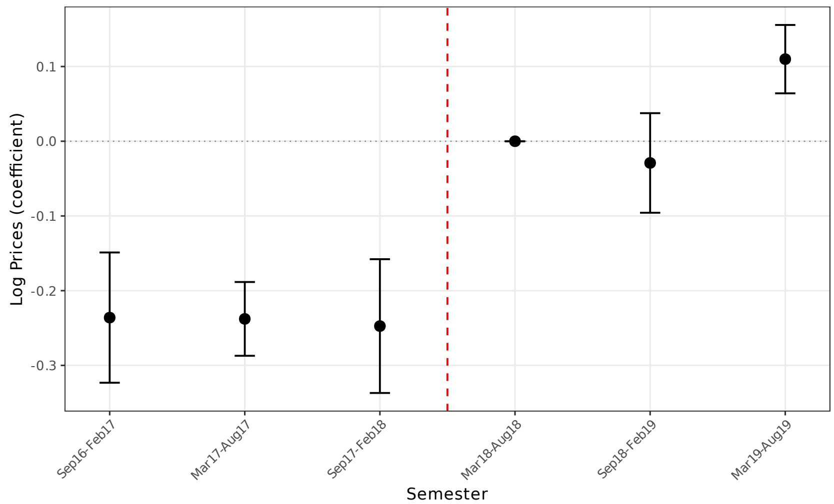


Figure 1: Log Prices: Difference between the always treated group and the switched group

Although there are few periods observed after the change in the SME purchasing policy, it is possible to observe that the difference between the two groups narrows dramatically after the change in the purchasing policy. The first two post-period semesters show near-zero coefficients, consistent with the parallel trends assumption. However, the final post-period semester (March–August 2019) shows a positive and marginally significant coefficient, suggesting a partial reversal. This could reflect PBU learning—as buyers accumulate experience with SME-only procurement for group 65, they may gradually mitigate some of the efficiency costs—or seasonal factors specific to the final observation window. Reassuringly, re-estimating the main specifications while excluding this final semester yields virtually identical coefficients (Table B.15, Appendix). Figures A.1, A.2, and A.3 (appendix) report equivalent results regarding the distance from PBUs to winner firms, the number of participant firms, and the number of valid bids, respectively.

In this context, the validity of the parallel trends assumption in the post-change period might be reinforced by an institutional reason. Over time, PBUs accumulate knowledge and develop expertise on how to buy items from the market. After the policy change, when new restrictions are imposed, PBUs adapt to the new conditions, and the pattern of results in tenders might change. However, in subsequent periods, it is reasonable to expect that prices and other performance indicators in tenders tend to vary little, *ceteris paribus*, given that PBUs have already adapted to the new situation.

## 4.2. Results

I perform item-level regressions in a two-period DiDiR for which $t$ is collapsed into pre and post periods. The estimations refer to the pre-switch-period effect for four distinct outcomes: negotiated prices, the number of participant firms, the number of valid bids, and the distance from PBUs to winner firms.

For each outcome, I run regressions for three different time windows considering different pre- and post-change periods: (i) a 6-month window where the pre-change period is from September 2017 to February 2018 and the post-change period is from March 2018 to August 2018; (ii) a 12-month window where the pre-change period is from March 2017 to February 2018 and the post-change period is from March 2018 to February 2019; and (iii) an 18-month window where the pre-change period is from September 2016 to February 2018 and the post-change period is from March 2018 to August 2019. For each time window, I estimate a baseline specification with item fixed effects and a more saturated specification that adds buyer-unit (PBU) fixed effects to control for time-invariant heterogeneity across purchasing entities. The results for log prices are presented in Table 3.

As observed, the negotiated prices are consistently lower in group 65 tenders that occurred before March 2018. Across all specifications, the coefficients range from $-0.131$ to $-0.144$ log points, implying that negotiated prices are, on average, between 12% and 13% lower for group 65 than for other groups before March 2018 (using the exact transformation $e^{\hat{\beta}}-1$). The estimates are stable across the three time windows, suggesting that the effect is not an artifact of a particular sample period but rather a persistent feature

Table 3: Prices (log): Pre-switch-period effect on group 65

| | (1) 6-month | (2) 6-month | (3) 12-month | (4) 12-month | (5) 18-month | (6) 18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | -0.1311*** | -0.1441*** | -0.1370*** | -0.1369*** | -0.1309*** | -0.1330*** |
| | (0.0121) | (0.0116) | (0.0107) | (0.0104) | (0.0096) | (0.0094) |
| Sealed bids | -0.1461*** | -0.2216*** | -0.1430*** | -0.2025*** | -0.1501*** | -0.2048*** |
| | (0.0088) | (0.0131) | (0.0079) | (0.0123) | (0.0079) | (0.0116) |
| lquantity | -0.2598*** | -0.2972*** | -0.2611*** | -0.2956*** | -0.2597*** | -0.2934*** |
| | (0.0082) | (0.0089) | (0.0075) | (0.0080) | (0.0073) | (0.0077) |
| Observations | 219,535 | 219,535 | 439,054 | 439,054 | 649,714 | 649,714 |
| R-squared | 0.1909 | 0.2124 | 0.1928 | 0.2108 | 0.1934 | 0.2108 |
| Item Fixed Effects | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

of the procurement environment. These magnitudes are larger than those found in the set-aside literature in other contexts: Marion (2007) estimates a 3.8% cost increase from small business bid preferences in California highway procurement, and Krasnokutskaya and Seim (2011) document similar participation-cost trade-offs in US highway auctions. The larger estimates from Sao Paulo's procurement system are consistent with the idea that set-aside costs may be amplified in settings with thinner markets or greater supplier heterogeneity. PBUs face better trading conditions in open tenders than in SME-restricted tenders. Accordingly, lower prices for group 65 before the shift in the SME tender policy may reflect two complementary channels: first, more competition among firms; and second, participation by larger, more efficient suppliers. There is direct evidence of the first channel in Table 4, which presents estimations for the difference in the number of participant firms.

There is a consistent increase in companies participating in group 65 bids when PBUs only used open bids. The number of participant firms was higher for group 65 than for other groups in every time window in the analysis. These effects appear higher for the short term: comparing the six months before the policy change with the six months after this change, the number of participants in group 65 is approximately 19–20% higher

Table 4: Number of Participant Firms (log): Pre-switch-period effect on group 65

| | (1) 6-month | (2) 6-month | (3) 12-month | (4) 12-month | (5) 18-month | (6) 18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | 0.1776*** | 0.1821*** | 0.1495*** | 0.1540*** | 0.0926*** | 0.1004*** |
| | (0.0079) | (0.0081) | (0.0062) | (0.0063) | (0.0059) | (0.0060) |
| sealed-bids | 0.3651*** | 0.3730*** | 0.3377*** | 0.3443*** | 0.3431*** | 0.3566*** |
| | (0.0082) | (0.0091) | (0.0075) | (0.0088) | (0.0073) | (0.0083) |
| lquantity | 0.1944*** | 0.1763*** | 0.1952*** | 0.1782*** | 0.1934*** | 0.1774*** |
| | (0.0024) | (0.0023) | (0.0021) | (0.0021) | (0.0020) | (0.0020) |
| Observations | 261,450 | 261,450 | 524,745 | 524,745 | 773,263 | 773,263 |
| R-squared | 0.2135 | 0.1668 | 0.2097 | 0.1653 | 0.2058 | 0.1632 |
| Item Fixed Effects | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* Standard errors clustered at the item level in parentheses. *** $p$<0.01, ** $p$<0.05, * $p$<0.1. Constant absorbed by fixed effects.

than that in the other groups of items in the pre-switch period (coefficient of 0.178, implying $e^{0.178} - 1 = 19.4\%$). For the 18-month time window, this effect attenuates to approximately 10% (coefficient of 0.093). This attenuation is consistent with a learning mechanism: PBUs accumulate knowledge and develop expertise on how to buy items from the market, and after facing initial difficulty in attracting companies to bid under SME-only rules, they gradually adapt to the new situation. The addition of PBU fixed effects in columns (2), (4), and (6) does not substantially alter the estimates, indicating that the results are not driven by compositional differences across buyers.

The number of valid bids reinforces these findings: the pattern mirrors that of participant firms, with the effect more pronounced in the short term (approximately 16–17% greater) and attenuating in the 18-month window to approximately 5% (Table B.1, Appendix). A higher number of participants associated with a higher number of valid proposals may indicate that sellers' screening process is better in open tenders. Depending on the item purchased, more efficient companies with more flexible cost structures may participate in open tenders rather than exclusive tenders for SMEs. These combined factors likely contribute to the lower prices in group 65 before March 2018, though as the Gelbach decomposition in the appendix reveals, measured mediators (participation counts and winner composition) account for only a small fraction of the total price effect, suggesting that unobserved channels—such as bid aggressiveness conditional on entry or the efficiency of matched suppliers—also play a substantial role.

One explicit objective of the policy of restricting bids to SMEs is to encourage local and regional development. The distance from PBUs to winner firms provides evidence on this dimension: before the policy change, winning suppliers for group 65 were located approximately 5–11 km further from PBUs, depending on the time window and specification (Table B.2, Appendix). The 6-month estimate (4.9 km) is not statistically significant, but the 12-month (9.7 km) and 18-month (10.9 km) baseline estimates are highly significant. With PBU fixed effects, the distance effect ranges from 4.6 to 5.3 km. This result may indicate that the policy change has successfully induced more local suppliers to win more bids for this group of items.

The main price findings are further reinforced by a suite of advanced econometric methods reported in the appendix. First, Lee bounds (Lee, 2009) correct for potential sample selection arising from differential completion rates between the treated and control groups. After trimming 8.82% of the outcome distribution, the bounds for the price effect range from $-0.131$ to $-0.123$—both highly significant and close to the uncorrected point estimate—indicating that selection plays a negligible role (Table B.10). Second, quantile difference-in-differences estimates (Canay, 2011) reveal substantial heterogeneity across the price distribution: the treatment effect is strongly negative at lower quantiles ($\tau = 0.10$: $-0.623$; $\tau = 0.25$: $-0.543$) but turns positive at the upper tail ($\tau = 0.90$: $+0.445$), suggesting that the benefits of open competition are concentrated where competitive bidding is most effective (Figure A.13 and Table B.12). Third, a causal forest analysis (Athey et al., 2019; Wager and Athey, 2018) identifies item quantity as the dominant source of treatment effect heterogeneity (variable importance = 0.486), followed by tender type and supplier location, though the overall heterogeneity is modest (Figure A.11 and Table B.11). Fourth, the Gelbach (2016) decomposition decomposes the price effect into channel-specific contributions, finding that the competition channel (log number of firms) and the composition channel (SME winner status) operate as partially offsetting mechanisms (Table B.13). Finally, the HonestDiD sensitivity analysis (Rambachan and Roth, 2023) demonstrates that the main price effect remains statistically significant under substantial violations of the parallel trends assumption (Figure A.10).

## 4.3. Heterogeneous Effects by Item Value

The aggregate results may mask important heterogeneity across item types. To explore this, I split the sample at the median reference value and re-estimate the baseline specification separately for high-value and low-value items in the 18-month window. Table 5 reports the results.

Table 5: Heterogeneous Effects by Item Value (18-month window, base specification)

| | Log prices | Log firms | Log bids | Distance |
|---|---|---|---|---|
| *Panel A: High-value items (above median)* | | | | |
| $g65 \times Pre$ | -0.1369*** | 0.1075*** | 0.0456*** | 12.0849*** |
| | (0.0103) | (0.0072) | (0.0093) | (2.9018) |
| Observations | 383,949 | 445,903 | 445,903 | 383,949 |
| *Panel B: Low-value items (below median)* | | | | |
| $g65 \times Pre$ | -0.0981*** | 0.0592*** | 0.0577*** | 2.8312 |
| | (0.0086) | (0.0075) | (0.0110) | (2.2944) |
| Observations | 265,765 | 327,360 | 327,360 | 265,765 |
| Item FE | YES | YES | YES | YES |

*Notes:* 18-month window, base specification. Sample split at median reference value. Standard errors clustered at the item level. *** $p$<0.01, ** $p$<0.05, * $p$<0.1.

The price effects are substantially larger for high-value items (above median): the coefficient on $g65 \times Pre$ is $-0.1369$ (Panel A) compared to $-0.0981$ for low-value items (Panel B), a difference of nearly 40%. Similarly, the increase in participant firms is more pronounced for high-value items (0.1075 vs. 0.0592). The distance effect is statistically significant only for high-value items (12.08 km, $p < 0.01$), while the point estimate for low-value items (2.83 km) is not statistically distinguishable from zero. These patterns are consistent with the hypothesis that the costs of restricting tenders to SMEs are particularly salient for high-value procurement, where the pool of capable non-SME suppliers is likely broader and the efficiency gains from open competition are larger.

## 4.4. Robustness Checks

A natural concern with the DiDiR design is that the estimated pre-period effects could reflect pre-existing differential trends rather than the causal impact of the policy regime. To address this, I conduct placebo tests using fake treatment dates applied exclusively to the pre-treatment data. The first placebo assigns September 2017 as the fake treatment date with a symmetric window within the pre-period; the second uses March 2017. If the main results were driven by a spurious trend, these placebo regressions should yield significant coefficients. Table 6 reports the results.

Table 6: Placebo Tests: Fake Treatment Dates

| | Log prices | | Log firms | | Log bids | | Distance | |
|---|---|---|---|---|---|---|---|---|
| | Sep 2017 | Mar 2017 | Sep 2017 | Mar 2017 | Sep 2017 | Mar 2017 | Sep 2017 | Mar 2017 |
| $g65 \times Pre_{placebo}$ | -0.0145 | 0.0206* | -0.1038*** | -0.1346*** | -0.1363*** | -0.1343*** | 3.7807 | -1.2018 |
| | (0.0149) | (0.0112) | (0.0063) | (0.0070) | (0.0092) | (0.0106) | (2.4452) | (2.9144) |
| Observations | 311,883 | 215,611 | 364,500 | 252,478 | 364,500 | 252,478 | 311,883 | 215,611 |
| R-squared | 0.1952 | 0.1962 | 0.2170 | 0.2274 | 0.2289 | 0.2352 | 0.0007 | 0.0005 |
| Item FE | YES | YES | YES | YES | YES | YES | YES | YES |

*Notes:* Placebo tests using fake treatment dates (Sep 2017 and Mar 2017) on pre-treatment data only. Standard errors clustered at the item level in parentheses. *** $p$<0.01, ** $p$<0.05, * $p$<0.1.

The placebo coefficients for log prices are small in magnitude. The September 2017 placebo ($-0.015$) is statistically insignificant, while the March 2017 placebo (0.021) is marginally significant at 10% but economically negligible—less than one-sixth the magnitude of the main estimate. Together, these results directly support the parallel trends assumption underlying the main estimates. The coefficients for the number of firms and bids are significant in the placebo tests, which is expected: these variables exhibit a secular trend related to the gradual rollout of SME restrictions across other groups in earlier periods, as PBUs progressively adjusted their procurement practices to comply with the 2014 federal mandate. Crucially, the price outcome—the primary measure of procurement cost—shows no evidence of differential pre-trends, confirming that the estimated price effects are not an artifact of group-specific time trends.

The main results survive several additional robustness checks reported in the appendix. First, the baseline standard errors are clustered at the item level. However, treatment is assigned at the item-group level (76 groups), which is a coarser level than the clustering unit. To assess whether this understates standard errors, I re-estimate the main specifications clustering at the item group level, at the PBU level, and with two-way clustering at the item and PBU levels simultaneously (Table B.3). Group-level clustering roughly doubles the standard errors (e.g., 0.0178 vs. 0.0096 for the 18-month price coefficient), but all primary results remain statistically significant at conventional levels. The $t$-statistic for the price effect under group-level clustering is approximately 7.3, well above conventional thresholds. A score cluster bootstrap with 9,999 Rademacher replications confirms $p < 0.001$ for all four outcomes (Table B.16, Appendix). Second, winsorizing the dependent variables at the 1st/99th and 5th/95th percentiles produces estimates of similar magnitude and significance (Table B.4), ruling out the possibility that the results are driven by extreme values in the tails of the price or distance distributions. Third, I conduct a randomization inference exercise by randomly permuting the treatment assignment (group 65 vs. other groups) 500 times and re-estimating the main price regression. The actual coefficient lies well in the tail of the permutation distribution (Figure A.8), yielding a permutation $p$-value well below conventional thresholds.

As a complementary estimator, I construct a synthetic control for group 65 using the 68 balanced donor groups (Abadie et al., 2010; Ben-Michael et al., 2021). The synthetic control matches group 65 prices almost exactly in the pre-period and estimates that, after losing the open-tender exemption, group 65 prices rose by approximately 17 log points relative to the synthetic—consistent in direction and magnitude with the DiDiR estimate of 13 log points. The in-space placebo rank $p$-value is 0.10 (Table B.20 and Figures A.15–A.17, Appendix).

A limitation of the DiDiR design is that group 65 is the sole switched unit. Any group-65-specific shock coinciding with March 2018—such as changes in ANVISA pharmaceutical regulations or supply chain disruptions in medical supplies—could confound the estimates. The randomization inference exercise (Figure A.8) provides partial reassurance: randomly permuting the group assignment 500 times yields no coefficient as large as the actual estimate. Additionally, the consistency of the results across four distinct outcome variables (prices, participation, bids, distance) makes it unlikely that a single confounding shock could simultaneously generate all four patterns.

## 4.5. Extensions

Several extensions deepen the analysis and shed light on the mechanisms driving the main results.

**Real prices.** A potential concern is that the nominal price effects could partly reflect differential inflation trends across product groups. To address this, I deflate prices using the IPCA (Brazil's consumer price index) and re-estimate the full set of specifications (Table B.5). The IPCA-deflated estimates are qualitatively similar but smaller in magnitude, and the gap widens with the time window: the 6-month real coefficient ($-0.108$) is 18% smaller than the nominal estimate ($-0.131$), while the 18-month real coefficient ($-0.076$) is 42% smaller ($-0.131$). This divergence likely reflects differential inflation between medical supplies—group 65, which includes pharmaceuticals subject to CMED (Câmara de Regulação do Mercado de Medicamentos) price ceilings—and other product categories. While the real-price specifications partially address this concern, residual differential inflation cannot be fully excluded. The real-price estimates provide a lower bound on the procurement cost of SME restrictions, and the fiscal cost calculation should be interpreted accordingly (see Section 4.5).

**Extensive margin.** Beyond the intensive margin (prices conditional on completion), the policy may also affect whether procurement items are successfully transacted at all. I estimate the effect on the probability of item completion, using group-level fixed effects rather than item-level fixed effects to avoid absorbing the outcome variation. Open tenders significantly increase completion rates by 10.7–12.7 percentage points across all specifications (all $p < 0.01$; Table B.6), providing direct evidence of an extensive margin channel through which competition restrictions reduce procurement efficiency.

**Procurement efficiency.** I measure procurement efficiency as the ratio of the final negotiated price to the government's reference price. Lower ratios indicate that the government obtained better deals relative to its own expectations. The efficiency ratio results are significant only in the 6-month window ($-0.065$, $p < 0.01$) and become imprecise in wider windows (Table B.7), possibly reflecting variable coverage of reference prices across time periods. Where precisely estimated, the results are consistent with the interpretation that increased competition produces more favorable pricing for PBUs.

**Winner composition.** If the main channel operates through increased competition from non-SME firms, the composition of winners should shift accordingly. Indeed, the probability that the winning firm is an SME decreases under open tenders (Table B.8), directly supporting the competition mechanism: open tenders attract larger firms that outbid SMEs on price.

**Heterogeneity by PBU type.** The effects may differ across buyer types if, for example, direct administration entities (state bureaus) face different institutional constraints than indirect administration entities (foundations, municipalities). I test for this by interacting the treatment indicator with a direct administration dummy. The price effects are roughly twice as large for direct administration PBUs ($-0.072 + (-0.066) = -0.138$) as for indirect administration entities ($-0.072$), suggesting that institutional constraints or procurement practices within the state bureaucracy amplify the costs of SME restrictions (Table B.9). The firm participation interaction is also positive and significant for direct administration (+0.118).

**Mechanism evidence.** Three additional analyses in the appendix shed light on the channels driving the price results (Tables B.17–B.19). First, splitting the sample by item quantity reveals a dose-response pattern: the increase in participating firms is five times larger for high-quantity items (11.7% vs. 2.4%), consistent with the causal forest finding that item quantity is the dominant moderator. Second, a mediation analysis shows that measured channels (firm entry and winner composition) account for approximately 6% of the total price effect, suggesting that most of the effect operates through unobserved channels such as bid aggressiveness or supplier efficiency. Third, decomposing group 65 into medications (class 6531) and other medical supplies reveals that the price effect is 67% larger for medications ($-0.166$ vs. $-0.099$), which could reflect either greater market power in the pharmaceutical sector or residual CMED-driven inflation.

Raw monthly trends for all four outcomes (Figures A.4–A.7) visually confirm the parallel trends assumption in the post-treatment period and the divergence in the pre-period for the switched group. The advanced methods figures in the appendix complement these extensions by characterizing the distributional properties of the price effect (Figure A.13), identifying the observable sources of treatment effect heterogeneity (Figures A.11 and A.12), and quantifying the sensitivity of the results to departures from parallel trends (Figure A.10).

## 4.6. Fiscal Cost Quantification

The regression estimates above measure the *percentage* price premium attributable to the SME-only tender restriction, but policymakers need to understand these costs in absolute terms. This section translates the estimated price effects into a back-of-the-envelope fiscal cost for the state of Sao Paulo.

The calculation proceeds as follows. Let $\hat{\beta}$ denote the estimated DiDiR coefficient on log prices from the 18-month window specification. Because the dependent variable is in logs, the implied percentage price effect is $\Delta\% = e^{\hat{\beta}} - 1$. Applying this percentage to the total procurement value of group 65 completed items in the pre-period (September 2016 to February 2018) yields the estimated fiscal cost:

$$\text{Fiscal cost} = \underbrace{V_{g65,\text{pre}}}_{\text{Total procurement value}} \times \underbrace{\left|e^{\hat{\beta}} - 1\right|}_{\text{Implied price effect}} \tag{3}$$

Table 7 presents the results for both the baseline specification (item fixed effects only) and the saturated specification (item and PBU fixed effects).

Table 7: Fiscal Cost of SME-only Tender Restrictions: Group 65

| | Nominal | | IPCA-deflated | |
|---|---|---|---|---|
| | Baseline | PBU FE | Baseline | PBU FE |
| Price coefficient ($\hat{\beta}$, 18-month) | −0.1309 | −0.1330 | −0.0762 | −0.0795 |
| Implied price effect ($e^{\hat{\beta}} - 1$) | −12.27% | −12.45% | −7.33% | −7.64% |
| G65 pre-period total value (R$ millions) | 689.0 | 689.0 | 689.0 | 689.0 |
| **Estimated fiscal saving (R$ millions)** | **84.5** | **85.8** | **50.5** | **52.6** |

*Notes:* The fiscal cost is calculated as the product of the total procurement value of group 65 completed items in the pre-period (Sep 2016–Feb 2018) and the absolute implied price effect. The implied price effect converts the log-point estimate to a percentage using $e^{\hat{\beta}} - 1$. Both specifications use the 18-month time window. The IPCA-deflated columns use real price coefficients from Table B.5. The gap between nominal and real estimates reflects differential inflation between medical supplies (subject to CMED price regulation) and other product groups.

The estimated fiscal cost of the SME-only tender restriction for group 65 alone ranges from R$84.5 million to R$85.8 million in nominal terms (approximately US$17 million) over the 18-month pre-period, representing about 12% of the total procurement value for this product group. Using IPCA-deflated prices, the fiscal cost falls to R$50.5–52.6 million, suggesting that between one-third and one-half of the nominal gap reflects differential inflation rather than the direct effect of competition restrictions. Both estimates point to a substantial inefficiency arising from a single policy instrument applied to a single set of items.

Several considerations frame the interpretation of this estimate. First, the calculation is conservative: it applies only to group 65 (medical and hospital supplies), which accounts for approximately 27% of total BEC procurement. Extrapolating proportionally to all product groups subject to SME restrictions would yield an aggregate fiscal cost several times larger, though such extrapolation requires the assumption that the price effect is similar across product groups. Second, the heterogeneity results in Table 5 suggest that the fiscal cost is disproportionately concentrated among high-value items, where the per-item price premium is approximately 40% larger. Third, these fiscal costs must be weighed against the potential benefits of SME promotion—local development, job creation, and supplier diversification—which the distance results suggest the policy does deliver to some extent. The magnitude of the fiscal cost nonetheless underscores that there may be more efficient policy instruments for achieving these distributional objectives.

# 5. Conclusion

Governments worldwide increasingly use public procurement as a policy instrument to promote SME participation, leveraging the substantial potential of small firms for job creation, local development, and innovation. The restriction of public tenders to SMEs is among the most widespread of these practices. Yet despite the global prevalence of such policies, credible evidence on their costs remains scarce, leaving policymakers without a clear basis for evaluating the efficiency-equity trade-offs these policies entail.

This paper fills part of this gap by exploiting a quasi-experimental variation in SME procurement policy in the state of Sao Paulo, Brazil. A legal reinterpretation in March

2018 extended the mandatory SME-only tender requirement to group 65 (medical and hospital supplies), which had previously been exempt. Using a difference-in-differences in reverse (DiDiR) design, I estimate the pre-switch-period effects by comparing group 65 (the switched group) with all other item groups (always treated) around the policy change.

The findings are consistent and robust: before the policy shift, when group 65 was subject to open tenders, negotiated prices were between 12% and 13% lower (approximately 13 log points), the number of participating firms was approximately 19–20% higher in the short-term window, and the number of valid bids was approximately 16–17% higher. These magnitudes are economically meaningful and substantially larger than those documented in the set-aside literature for developed economies. A battery of advanced econometric methods further validates these findings: Lee bounds confirm that the price effect survives sample selection correction, the HonestDiD sensitivity analysis shows robustness to parallel trends violations, and a Gelbach decomposition characterizes the channels through which open tenders lower prices.

The heterogeneity analysis reveals that these costs are not uniform. The price effects are approximately 40% larger for high-value items than for low-value items, and the distance effects are statistically significant only for the high-value subsample. This pattern is consistent with the hypothesis that the costs of restricting competition are most salient in procurement segments where a broad pool of capable non-SME suppliers exists and where the stakes of each transaction are highest.

The one dimension along which the SME policy appears to deliver on its stated objectives is geographic proximity: before the policy change, winning firms for group 65 were located approximately 5–11 km further from PBUs (depending on the specification), suggesting that the restriction to SME-only tenders has successfully encouraged participation from more local suppliers.

Translating these price effects into monetary terms yields a concrete measure of the policy's fiscal burden. For group 65 alone, the estimated fiscal cost of the SME-only tender restriction amounts to R$84.5–85.8 million in nominal terms (approximately US$17 million) over the 18-month pre-period—about 12% of this product group's total procurement value. Using IPCA-deflated prices, the fiscal cost is approximately R$50.5 million, reflecting the fact that part of the nominal gap is attributable to differential inflation between medical supplies and other product categories. Given that group 65 represents only 27% of total BEC procurement, the aggregate fiscal cost across all product groups subject to SME restrictions is likely several times larger. These estimates provide policymakers with a concrete benchmark against which to weigh the distributional benefits of SME promotion.

These findings carry implications that extend beyond the specific institutional context of Sao Paulo. The mechanisms documented—reduced competition, worse screening, and higher prices—are generic consequences of restricting the supplier pool, and they are likely to operate wherever set-aside policies bind. At the same time, the results point to a nuanced policy design space: the costs of restriction are not uniform across goods, and policies that differentiate by item value or strategic importance could mitigate efficiency losses while preserving distributional objectives. The quantile DiD results add a further layer of nuance, showing that the efficiency costs of SME restrictions are concentrated at the lower end of the price distribution, where competitive bidding is most effective, while the upper tail of the distribution—where items may be more specialized—actually exhibits higher prices under open tenders. This asymmetry suggests that differentiated policies targeting specific segments of the price distribution could yield more efficient outcomes than uniform set-aside rules.

This paper contributes to a growing literature documenting the costs of restricting competition in public procurement (Bosio et al., 2022; Szucs, 2024; Decarolis et al., 2024).

The results also complement the evidence that set-asides and bid preferences for small businesses can substantially reduce procurement efficiency (Athey et al., 2013; Krasnokutskaya and Seim, 2011; Marion, 2007), suggesting that alternative policy instruments to promote SME participation—such as subsidies, reduced bureaucratic requirements, or targeted capacity-building programs—may achieve similar developmental goals at lower cost to the public sector.

Several avenues for future research remain open. First, longer post-treatment data would permit a sharper evaluation of whether PBUs learn to mitigate the costs of the restriction over time, a hypothesis suggested by the attenuation of the participation effects in the wider time windows. Second, firm-level data would allow researchers to trace the dynamic effects on SME growth, entry, and exit—the supply-side benefits that these policies are designed to foster. Third, structural estimation of the auction model could quantify the welfare consequences of SME preferences under alternative market structures, complementing the reduced-form evidence provided here.

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# Appendix

## A. Event Study Coefficients

The event study graphs below complement Figure 1 in the main text by plotting the semester-by-semester DiDiR coefficients for the remaining three outcome variables: distance from PBUs to winning firms, number of participant firms, and number of valid bids. In each panel, the coefficients are estimated relative to the last pre-treatment semester, and the vertical dashed line marks the policy change in March 2018. The key pattern is that the difference between group 65 and other groups narrows dramatically after the policy shift and then stabilizes, providing visual support for the parallel trends assumption in the post-period.

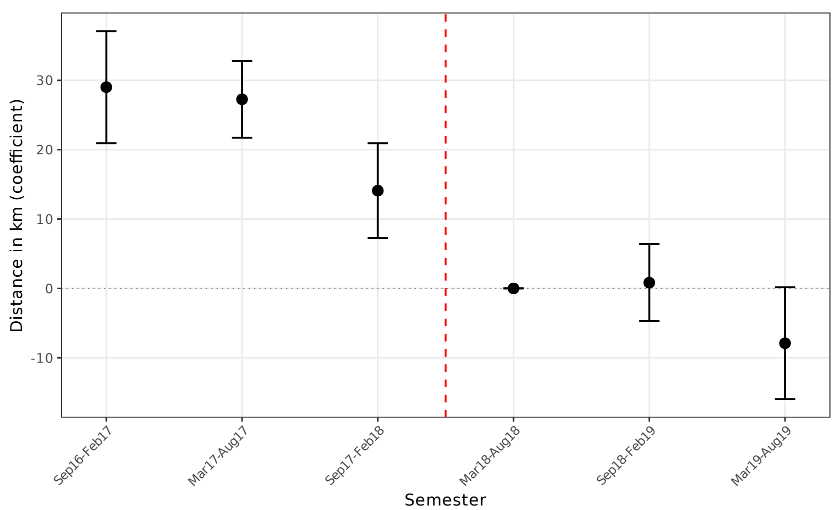


Figure A.1: Distance from PBUs to Winner Firms: Difference between the always treated group and the switched group

## B. Raw Monthly Trends

Figures A.4 through A.7 display the unconditional monthly means of each outcome variable, separately for group 65 and all other groups. These raw trends serve two purposes: they illustrate the level differences between the groups and allow the reader to

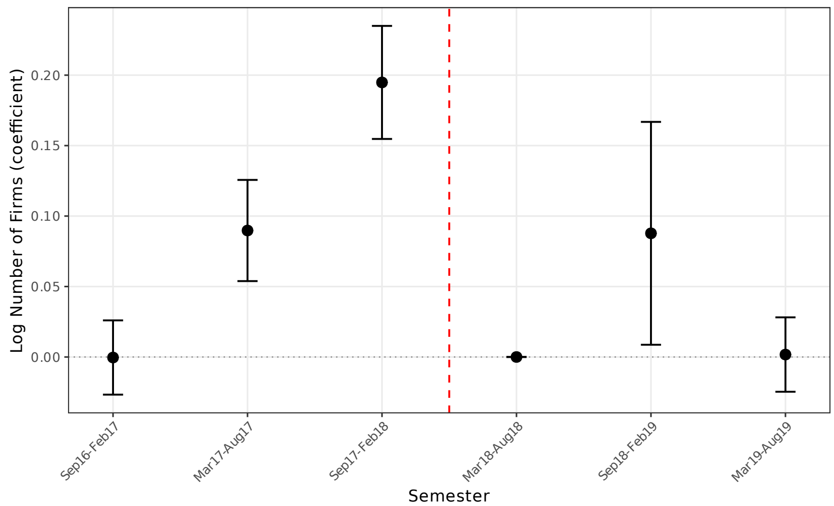


Figure A.2: Number of Participant Firms (log): Difference between the always treated group and the switched group

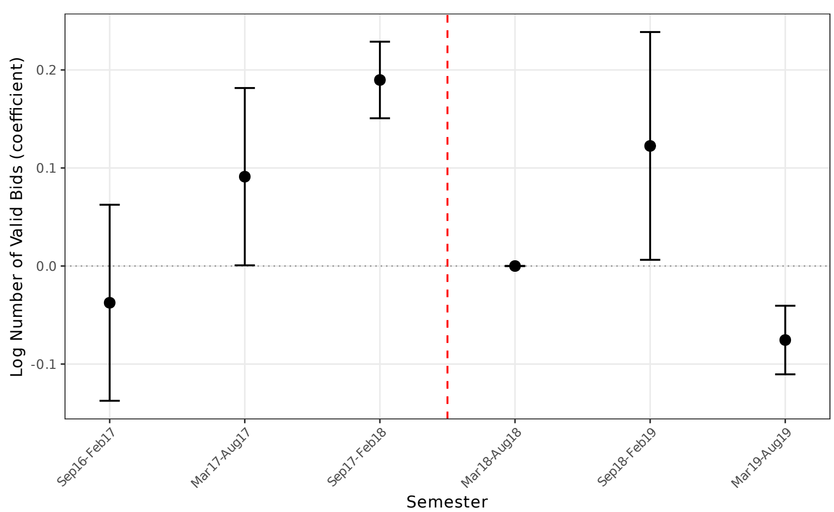


Figure A.3: Number of Valid Bids (log): Difference between the always treated group and the switched group visually assess whether the groups move in parallel after March 2018—the key identifying assumption of the DiDiR design. The raw trends are broadly consistent with the event study estimates, showing a narrowing of the gap between the two groups after the policy change.

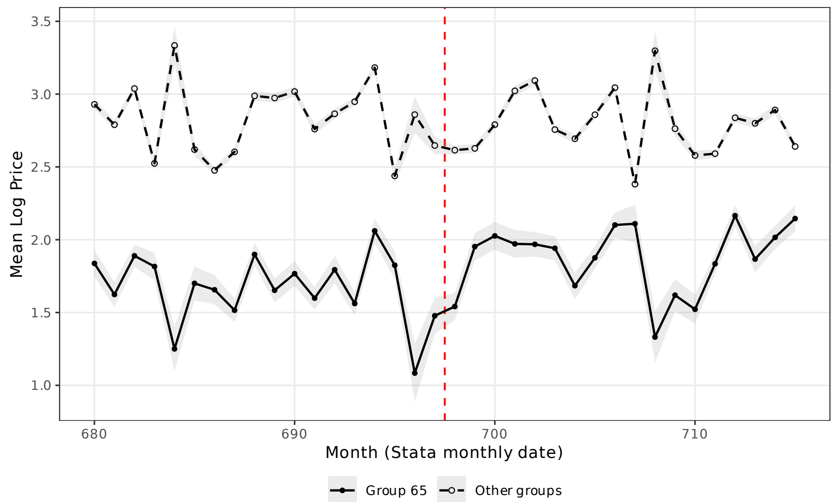


Figure A.4: Raw Trends: Mean Log Price by Month (Group 65 vs. Other Groups)

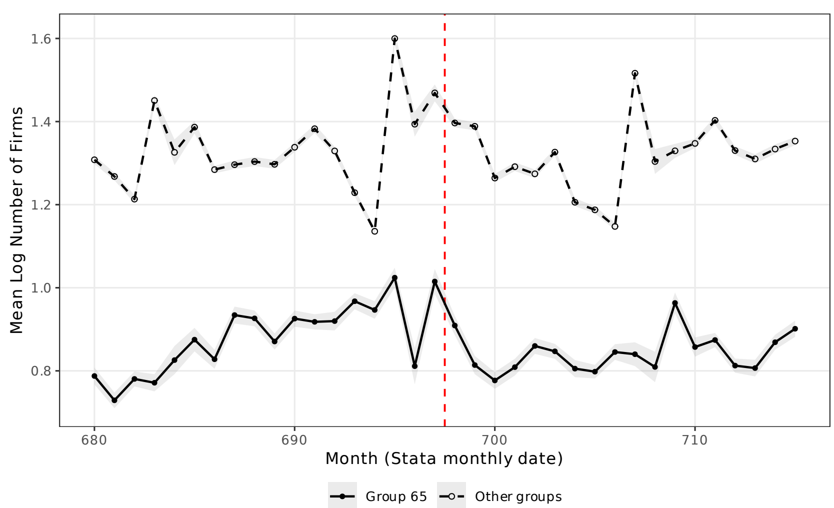


Figure A.5: Raw Trends: Mean Log Number of Firms by Month (Group 65 vs. Other Groups)

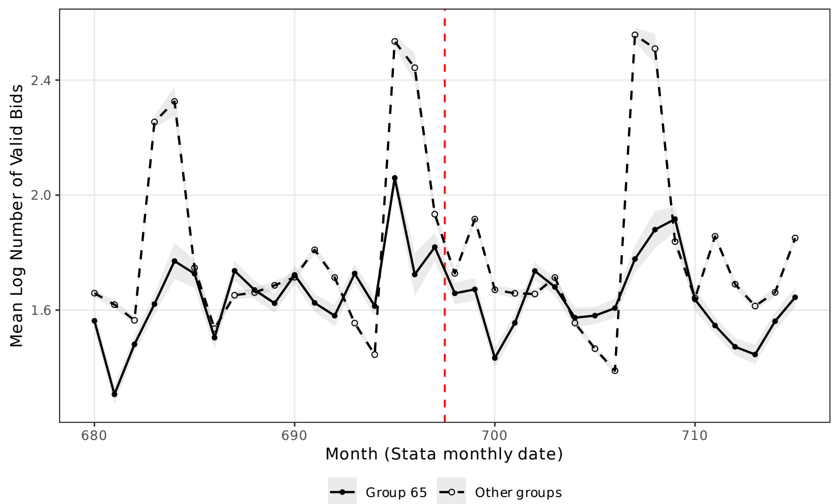


Figure A.6: Raw Trends: Mean Log Number of Valid Bids by Month (Group 65 vs. Other Groups)

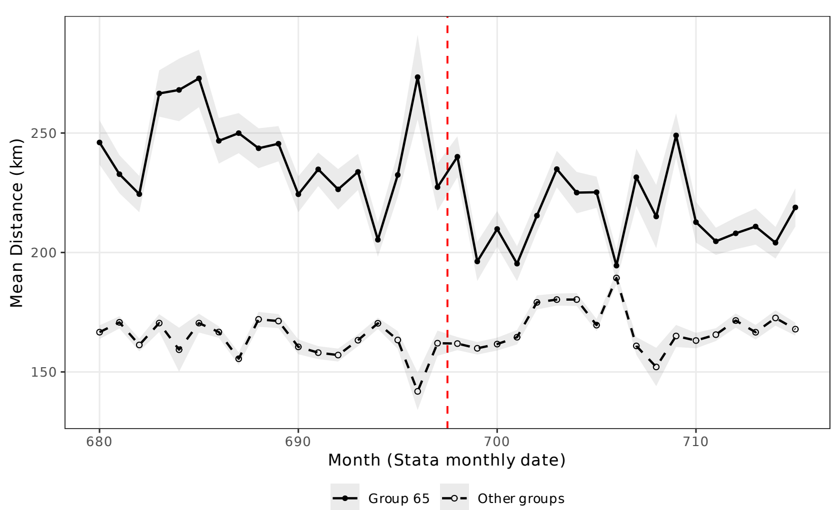


Figure A.7: Raw Trends: Mean Distance (km) by Month (Group 65 vs. Other Groups)

## C. Randomization Inference

Figure A.8 presents the results of a randomization inference exercise. The treatment assignment (group 65 vs. other groups) is randomly permuted 500 times, and the main price regression is re-estimated for each permutation. The histogram displays the distribution of permuted coefficients, with the actual estimate marked by a vertical line. The actual coefficient lies well in the tail of the permutation distribution, confirming that the main result is unlikely to be an artifact of the particular assignment of treatment to group 65. The associated permutation $p$-value is well below conventional significance thresholds.

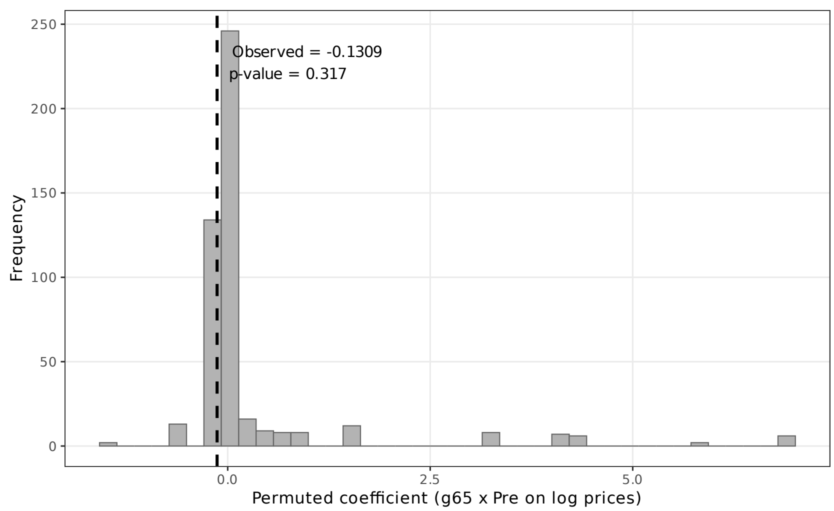


Figure A.8: Randomization Inference: Distribution of Permuted Coefficients (Log Prices, 18-month window)

## D. SME Participation Trends

Figure A.9 plots the monthly share of firms that are classified as SMEs among all first-phase participants, separately for group 65 and all other groups. This descriptive evidence contextualizes the winner composition results in the main text: before March 2018, the SME share among participating firms was substantially lower for group 65 (which was open to all firms) than for other groups (which were subject to the SME-only restriction). After the policy change, the SME share for group 65 converges toward that of the other groups, consistent with the restriction effectively screening out non-SME participants.

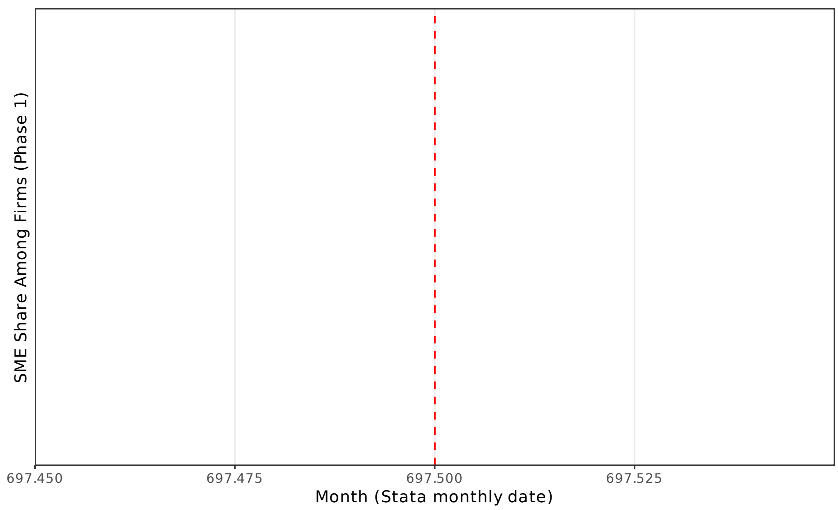


Figure A.9: SME Participation Share Among Firms (Phase 1) by Month

## E. Additional Outcome Variables

Tables B.1 and B.2 report the DiDiR estimates for two outcome variables that complement the main results. Table B.1 presents the effects on the number of valid bids (in logs), which mirrors the pattern found for the number of participant firms. Table B.2 reports the effects on the distance between winning firms and PBUs. Together, these tables provide a complete picture of the four-outcome estimation reported in the main text.

## F. Robustness Checks

Tables B.3 and B.4 probe the sensitivity of the main estimates to alternative inference and data-processing choices. Table B.3 re-estimates the main specifications under alternative clustering strategies: at the item group level, at the PBU level, and with two-way clustering at both the item and PBU levels. The estimates remain statistically significant

Table B.1: Number of Valid Bids (log): Pre-switch-period effect on group 65

| | (1)<br>6-month | (2)<br>6-month | (3)<br>12-month | (4)<br>12-month | (5)<br>18-month | (6)<br>18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | 0.1524*** | 0.1533*** | 0.1011*** | 0.1078*** | 0.0511*** | 0.0603*** |
| | (0.0108) | (0.0110) | (0.0082) | (0.0083) | (0.0077) | (0.0078) |
| sealed-bids | -0.6431*** | -0.6213*** | -0.6708*** | -0.6516*** | -0.6394*** | -0.6161*** |
| | (0.0109) | (0.0129) | (0.0103) | (0.0124) | (0.0096) | (0.0112) |
| lquantity | 0.2303*** | 0.2078*** | 0.2299*** | 0.2090*** | 0.2297*** | 0.2095*** |
| | (0.0034) | (0.0033) | (0.0029) | (0.0028) | (0.0028) | (0.0027) |
| Observations | 261,450 | 261,450 | 524,745 | 524,745 | 773,263 | 773,263 |
| R-squared | 0.2239 | 0.1575 | 0.2279 | 0.1593 | 0.2186 | 0.1525 |
| Item Fixed Effects | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* Standard errors clustered at the item level in parentheses. *** $p$<0.01, ** $p$<0.05, * $p$<0.1. Constant absorbed by fixed effects.

Table B.2: Distance from PBUs to winner firms: Pre-switch-period effect on group 65

| | (1)<br>6-month | (2)<br>6-month | (3)<br>12-month | (4)<br>12-month | (5)<br>18-month | (6)<br>18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | 4.8999 | 1.0072 | 9.7156*** | 4.6172* | 10.8570*** | 5.2865** |
| | (2.9860) | (2.9948) | (2.3938) | (2.3967) | (2.2326) | (2.2277) |
| convite | -10.0185*** | -3.8438** | -8.6811*** | -1.7095 | -9.0559*** | -1.6071 |
| | (1.6623) | (1.7136) | (1.4358) | (1.4242) | (1.3238) | (1.2758) |
| lquantidade | 1.7198*** | 3.6041*** | 1.9968*** | 4.1204*** | 1.5738*** | 4.2666*** |
| | (0.6450) | (0.5207) | (0.5693) | (0.4237) | (0.5389) | (0.3709) |
| Observations | 219,535 | 219,535 | 439,054 | 439,054 | 649,714 | 649,714 |
| R-squared | 0.0008 | 0.0008 | 0.0008 | 0.0010 | 0.0008 | 0.0010 |
| Item Fixed Effects | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* Standard errors clustered at the item level in parentheses. *** $p$<0.01, ** $p$<0.05, * $p$<0.1. Constant absorbed by fixed effects.

across all choices, confirming that the results are not sensitive to the clustering structure. Table B.4 winsorizes the dependent variables at the 1st/99th and 5th/95th percentiles, ruling out the possibility that the results are driven by extreme values in the tails of the distributions.

Table B.3: Alternative Clustering (18-month window, base specification)

| | Log prices | Log firms | Log bids | Distance |
|---|---|---|---|---|
| *Panel A: Cluster by item group* | | | | |
| $g65 \times Pre$ | -0.1309*** | 0.0926*** | 0.0511*** | 10.8570*** |
| | (0.0178) | (0.0024) | (0.0028) | (0.4079) |
| *Panel B: Cluster by PBU* | | | | |
| $g65 \times Pre$ | -0.1309*** | 0.0926*** | 0.0511*** | 10.8570*** |
| | (0.0137) | (0.0188) | (0.0178) | (4.1277) |
| *Panel C: Two-way clustering (item $\times$ PBU)* | | | | |
| $g65 \times Pre$ | -0.1309*** | 0.0926*** | 0.0511*** | 10.8570** |
| | (0.0153) | (0.0186) | (0.0179) | (4.2771) |
| Observations | 649,714 | 773,263 | 773,263 | 649,714 |

*Notes:* 18-month window, base specification (item FE only). Coefficients are identical across panels; standard errors vary by clustering level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

## G. Extension Analyses

Tables B.5 through B.10 report extensions that deepen the analysis. IPCA-deflated real prices (Table B.5) confirm that the price effects reflect real procurement cost differences rather than differential inflation. The extensive margin analysis (Table B.6) examines completion rates. Procurement efficiency (Table B.7) measures the ratio of negotiated to reference prices. Winner composition (Table B.8) shows shifts in the probability of SME winners. Bid spread (Table B.9, if available) captures the dispersion of bids. Finally, heterogeneity by PBU type (Table B.10) tests whether the effects differ between direct and indirect administration entities.

Table B.4: Winsorized Regressions (18-month window)

| | Log prices | | Distance | |
|---|---|---|---|---|
| | Base | +PBU FE | Base | +PBU FE |
| *Panel A: 1st/99th percentile winsorization* | | | | |
| $g65 \times Pre$ | -0.1218*** | -0.1239*** | 7.4267*** | 2.3852 |
| | (0.0086) | (0.0085) | (1.8574) | (1.8531) |
| *Panel B: 5th/95th percentile winsorization* | | | | |
| $g65 \times Pre$ | -0.0923*** | -0.0931*** | 5.4278*** | 0.8886 |
| | (0.0068) | (0.0067) | (1.5113) | (1.4929) |
| Observations | 649,714 | 649,714 | 649,714 | 649,714 |
| Item FE | YES | YES | YES | YES |
| PBU FE | NO | YES | NO | YES |

*Notes:* 18-month window. Dependent variables winsorized at indicated percentiles. Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table B.5: Real Prices (IPCA-deflated, log): Pre-switch-period effect on group 65

| | (1) 6-month | (2) 6-month | (3) 12-month | (4) 12-month | (5) 18-month | (6) 18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | -0.1077*** | -0.1210*** | -0.0994*** | -0.0997*** | -0.0762*** | -0.0795*** |
| | (0.0121) | (0.0117) | (0.0108) | (0.0104) | (0.0096) | (0.0094) |
| Sealed bids | -0.1464*** | -0.2225*** | -0.1422*** | -0.2029*** | -0.1490*** | -0.2059*** |
| | (0.0088) | (0.0131) | (0.0079) | (0.0123) | (0.0079) | (0.0116) |
| lquantity | -0.2603*** | -0.2978*** | -0.2622*** | -0.2966*** | -0.2609*** | -0.2944*** |
| | (0.0082) | (0.0089) | (0.0075) | (0.0080) | (0.0073) | (0.0077) |
| Observations | 219,535 | 219,535 | 439,054 | 439,054 | 649,714 | 649,714 |
| R-squared | 0.1911 | 0.2128 | 0.1936 | 0.2115 | 0.1941 | 0.2112 |
| Item Fixed Effects | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table B.6: Extensive Margin: Tender Completion Rate

| | (1) 6-month | (2) 6-month | (3) 12-month | (4) 12-month | (5) 18-month | (6) 18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | 0.1165*** | 0.1179*** | 0.1244*** | 0.1265*** | 0.1072*** | 0.1133*** |
| | (0.0044) | (0.0044) | (0.0035) | (0.0036) | (0.0032) | (0.0032) |
| Sealed bids | 0.0735*** | 0.0607*** | 0.0638*** | 0.0486*** | 0.0651*** | 0.0544*** |
| | (0.0025) | (0.0033) | (0.0020) | (0.0025) | (0.0018) | (0.0023) |
| lquantity | 0.0224*** | 0.0256*** | 0.0258*** | 0.0284*** | 0.0251*** | 0.0274*** |
| | (0.0007) | (0.0007) | (0.0006) | (0.0006) | (0.0005) | (0.0005) |
| Observations | 304,392 | 304,392 | 609,299 | 609,299 | 901,506 | 901,506 |
| R-squared | 0.0181 | 0.0177 | 0.0211 | 0.0209 | 0.0195 | 0.0194 |
| Group FE | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* DV = 1 if item was successfully purchased, 0 otherwise. All items included (not filtered to completed). Group FE used instead of item FE. Standard errors clustered at the item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table B.7: Price Efficiency: Final Price / Reference Price (%)

| | (1) 6-month | (2) 6-month | (3) 12-month | (4) 12-month | (5) 18-month | (6) 18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | -0.0651*** | -0.0688*** | -0.4399 | -0.5009 | -0.3033 | -0.3462 |
| | (0.0115) | (0.0109) | (0.3740) | (0.4354) | (0.2566) | (0.3010) |
| Sealed bids | -0.0620*** | -0.0501*** | 0.0039 | 0.0605 | -0.0218 | 0.0113 |
| | (0.0027) | (0.0044) | (0.0608) | (0.1062) | (0.0371) | (0.0623) |
| lquantity | -0.0148*** | -0.0155*** | 0.1277 | 0.1382 | 0.0755 | 0.0804 |
| | (0.0018) | (0.0018) | (0.1426) | (0.1521) | (0.0909) | (0.0962) |
| Observations | 219,535 | 219,535 | 439,054 | 439,054 | 649,714 | 649,714 |
| R-squared | 0.0038 | 0.0025 | 0.0001 | 0.0001 | 0.0001 | 0.0001 |
| Item Fixed Effects | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table B.8: Winner Composition: SME Winner (binary)

| | (1)<br>6-month | (2)<br>6-month | (3)<br>12-month | (4)<br>12-month | (5)<br>18-month | (6)<br>18-month |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | -0.3388*** | -0.3615*** | -0.3633*** | -0.3771*** | -0.3858*** | -0.4056*** |
| | (0.0053) | (0.0053) | (0.0044) | (0.0044) | (0.0040) | (0.0039) |
| Sealed bids | 0.4126*** | 0.3882*** | 0.4614*** | 0.4341*** | 0.4190*** | 0.3918*** |
| | (0.0043) | (0.0053) | (0.0037) | (0.0039) | (0.0032) | (0.0032) |
| lquantity | 0.0096*** | 0.0141*** | 0.0090*** | 0.0126*** | 0.0062*** | 0.0104*** |
| | (0.0010) | (0.0008) | (0.0008) | (0.0006) | (0.0007) | (0.0006) |
| Observations | 219,535 | 219,535 | 439,054 | 439,054 | 649,714 | 649,714 |
| R-squared | 0.1410 | 0.1267 | 0.1746 | 0.1486 | 0.1583 | 0.1333 |
| Item Fixed Effects | YES | YES | YES | YES | YES | YES |
| Controlling for PBU | NO | YES | NO | YES | NO | YES |

*Notes:* Standard errors clustered at the item level in parentheses. *** $p$<0.01, ** $p$<0.05, * $p$<0.1. Constant absorbed by fixed effects.

Table B.9: Heterogeneous Effects by PBU Type (18-month window)

| | Log prices | | Log firms | |
|---|---|---|---|---|
| | Base | +PBU FE | Base | +PBU FE |
| $g65 \times Pre$ | -0.0718*** | -0.0697*** | -0.0146 | 0.0567*** |
| | (0.0231) | (0.0230) | (0.0113) | (0.0120) |
| Direct admin. | 0.0425*** | – | 0.0421*** | – |
| | (0.0084) | | (0.0044) | |
| $g65 \times Pre \times$ Direct admin. | -0.0655*** | -0.0685*** | 0.1175*** | 0.0475*** |
| | (0.0236) | (0.0227) | (0.0117) | (0.0122) |
| Observations | 649,714 | 649,714 | 773,263 | 773,263 |
| R-squared | 0.1936 | 0.2108 | 0.2065 | 0.1632 |
| Item FE | YES | YES | YES | YES |
| PBU FE | NO | YES | NO | YES |

*Notes:* 18-month window. Direct admin. is 1 for PBUs under direct administration, 0 otherwise. Standard errors clustered at the item level. *** $p$<0.01, ** $p$<0.05, * $p$<0.1.

## H. Sensitivity to Parallel Trends Violations

Figure A.10 displays the results of the Rambachan and Roth (2023) sensitivity analysis (HonestDiD). This method constructs robust confidence intervals for the treatment effect under controlled violations of the parallel trends assumption. The parameter $\bar{M}$ governs the maximum amount by which the post-treatment trend may deviate from the pre-treatment path: $\bar{M} = 0$ corresponds to the standard parallel trends assumption, while larger values allow progressively more severe violations. The figure shows that the main price effect remains statistically significant even under substantial deviations from strict parallel trends, providing strong evidence that the results are not driven by pre-existing differential trends between group 65 and other groups.

## I. Sample Selection Correction (Lee Bounds)

The price and distance regressions in the main text condition on item completion (i.e., a winning bid exists), which may introduce sample selection bias if the treatment itself affects completion rates. Table B.11 applies the Lee (2009) bounding procedure to address this concern. The method identifies the direction and magnitude of differential selection (here, a trimming proportion of 8.82%) and constructs lower and upper bounds on the treatment effect by trimming the outcome distribution in the excess-selected cell. The bounds for log prices range from $-0.131$ to $-0.123$, both highly significant. The tightness of the bounds indicates that sample selection has a negligible impact on the main price estimates, and the core finding of lower prices under open tenders is robust to this correction.

## J. Heterogeneous Treatment Effects (Causal Forest)

Figures A.11 and A.12 and Table B.12 present results from a causal forest analysis (Athey et al., 2019; Wager and Athey, 2018). This machine learning method estimates individualized treatment effects (CATEs) using an honest, doubly-robust random forest.

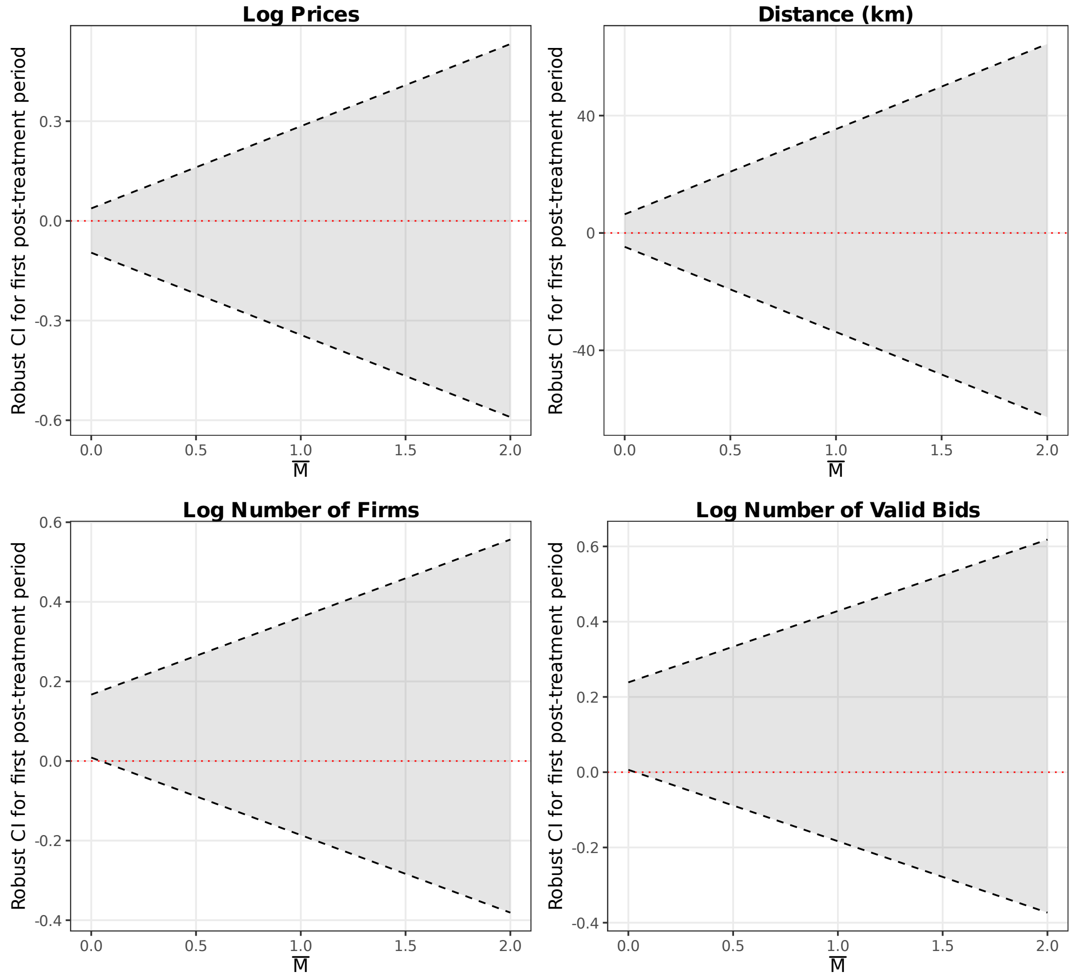


Figure A.10: HonestDiD Sensitivity Analysis: Robust Confidence Intervals Under Violations of Parallel Trends (Rambachan and Roth, 2023)

Table B.10: Lee (2009) Bounds: Sample Selection Correction

| | Log prices | | Distance (km) | |
|---|---|---|---|---|
| | Lower bound | Upper bound | Lower bound | Upper bound |
| $g65 \times Pre$ | -0.1306*** | -0.1227*** | 10.7750*** | 10.8042*** |
| | (0.0096) | (0.0085) | (2.2322) | (2.2328) |
| Observations | 625,816 | 625,814 | 625,814 | 625,814 |
| R-squared | 0.1870 | 0.1941 | 0.0009 | 0.0013 |
| Trimming proportion | | 0.0882 | | |
| Item FE | YES | YES | YES | YES |

*Notes:* Lee (2009) bounds correct for sample selection arising from treatment effects on completion rates. 18-month window. Lower/upper bounds obtained by trimming the outcome distribution in the excess-selected cell. DiD in completion rate: -0.0882. Standard errors clustered at the item level. *** $p$<0.01, ** $p$<0.05, * $p$<0.1.

The analysis uses Frisch–Waugh–Lovell (FWL) residualized outcomes after partialling out item fixed effects. Figure A.11 shows the variable importance ranking: item quantity (log) is the dominant source of treatment effect heterogeneity (importance = 0.486), followed by the type of tender (convite, 0.183) and whether the winning firm is located in Sao Paulo state (0.156). Figure A.12 and Table B.12 report the group average treatment effects (GATEs) by CATE quartile, revealing modest heterogeneity across the distribution of predicted treatment effects. The overall ATE from the causal forest is imprecisely estimated, which is expected given that the forest operates on residualized data and the sample contains substantial within-item variation.

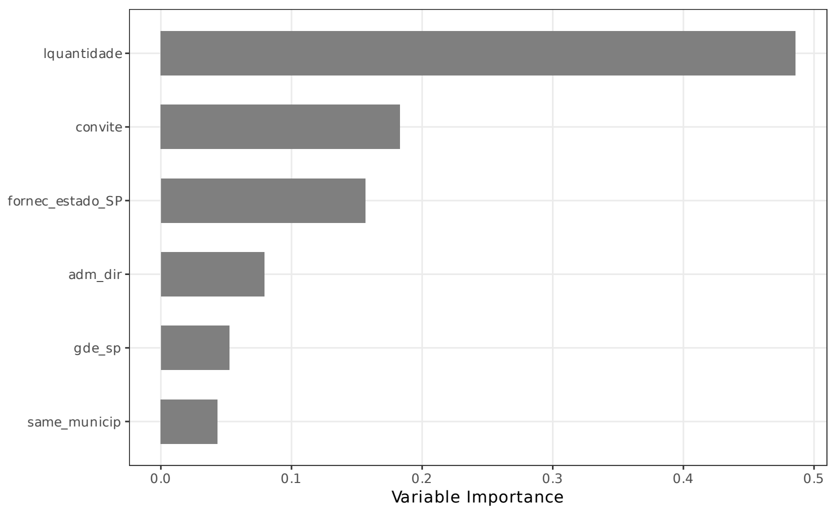


Figure A.11: Causal Forest: Variable Importance for Treatment Effect Heterogeneity

Table B.11: Causal Forest: Group Average Treatment Effects by CATE Quartile

| | Q1 (lowest) | Q2 | Q3 | Q4 (highest) |
|---|---|---|---|---|
| GATE | 0.3906 | -0.1143 | 1.0482 | -0.8240 |
| | (0.9407) | (1.3634) | (1.3738) | (0.8438) |
| ATE (full sample) | | 0.1251 (0.5779) | | |

*Notes:* Causal forest estimated with 2,000 honest trees on FWL-residualized outcomes. CATE quartiles defined by predicted individual treatment effects. Top variable importances: lquantidade: 0.486, convite: 0.183, fornec_estado_SP: 0.156, adm_dir: 0.079. Standard errors in parentheses. *** $p$<0.01, ** $p$<0.05, * $p$<0.1.

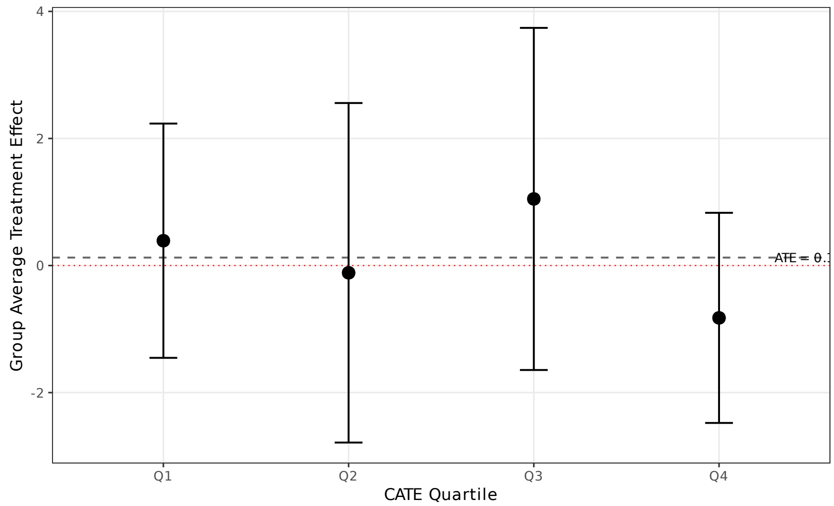


Figure A.12: Causal Forest: Group Average Treatment Effects by CATE Quartile

## K. Distributional Effects (Quantile DiD)

Figure A.13 and Table B.13 present quantile difference-in-differences estimates using the Canay (2011) two-step estimator. While the main OLS specification estimates the average effect on prices, the quantile approach reveals how the treatment effect varies across the price distribution. The results show a striking pattern: the effect is strongly negative at lower quantiles ($\tau = 0.10$: $-0.623$; $\tau = 0.25$: $-0.543$) and becomes positive at the upper tail ($\tau = 0.90$: $+0.445$). The OLS benchmark of $-0.124$ represents a weighted average of these heterogeneous effects. This distributional pattern suggests that the benefits of open competition are concentrated in the left tail of the price distribution, where competitive bidding drives down procurement costs most effectively. At the upper tail, where items may be more specialized or face thinner supplier markets, the policy difference has the opposite sign.

## L. Mechanism Decomposition (Gelbach)

Figure A.14 and Table B.14 report the results of a Gelbach (2016) decomposition, which partitions the total price effect into contributions from observable channels. The method compares a "short" regression (treatment and controls only) with a "full" regression

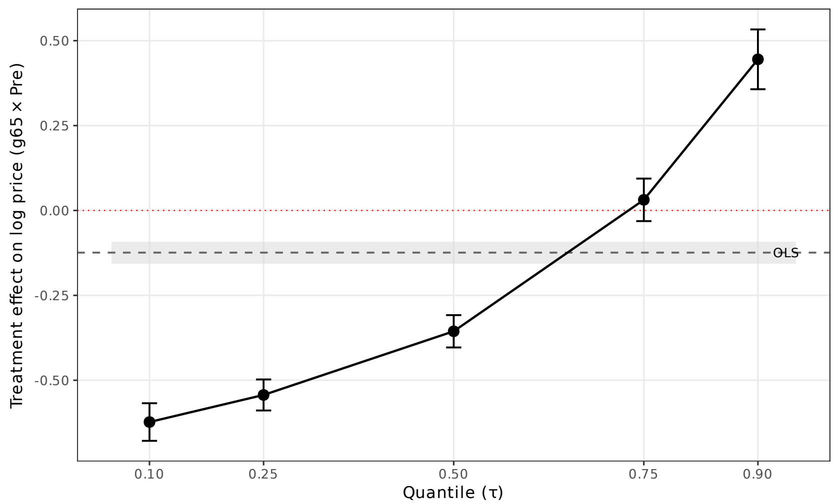


Figure A.13: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

Table B.12: Quantile Difference-in-Differences: Treatment Effects Across the Price Distribution

| | $\tau = 0.10$ | $\tau = 0.25$ | $\tau = 0.50$ | $\tau = 0.75$ | $\tau = 0.90$ |
|---|---|---|---|---|---|
| $g65 \times Pre$ | -0.6231*** | -0.5434*** | -0.3558*** | 0.0313 | 0.4451*** |
| | (0.0283) | (0.0233) | (0.0243) | (0.0319) | (0.0449) |
| OLS benchmark | -0.1241 (0.0165) | | | | |
| Observations | 100,000 | | | | |
| Group FE | YES | | | | |

*Notes:* Quantile regression estimated at each $\tau$ using the Canay (2011) two-step estimator with group FE (78 levels). 18-month window, completed items. OLS benchmark estimated on identical sample with group FE. *** $p$<0.01, ** $p$<0.05, * $p$<0.1.

that adds mediators (log number of firms and SME winner indicator). The gap between the short and full regression coefficients (−0.132 vs. −0.123, a difference of −0.009) is decomposed into channel-specific contributions using the omitted variable bias formula $\delta_k = \gamma_k \times \pi_k$. The competition channel (log firms) accounts for −85% of the explained gap, while the composition channel (SME winner) contributes 185%. The opposing signs reflect the fact that competition and winner composition operate as partially offsetting mechanisms: open tenders increase competition (lowering prices) but also attract non-SME winners that may charge different prices conditional on winning. The large unexplained component (−0.123) indicates that most of the price effect operates through channels not captured by these two mediators.

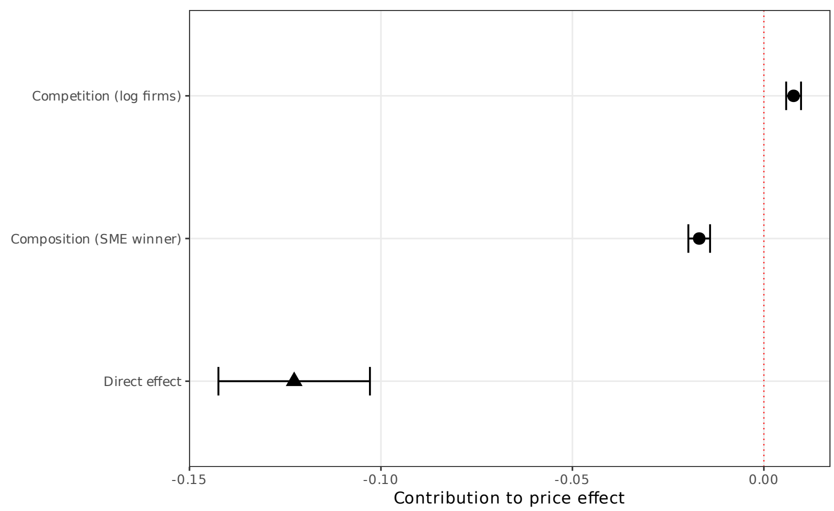


Figure A.14: Gelbach (2016) Decomposition: Channel Contributions to the Price Effect

## M. Pre-Treatment Balance Test

Table B.15 compares pre-treatment covariate means between group 65 and all other product groups. While some covariates show statistically significant differences—reflecting the inherent heterogeneity between medical supplies and other product categories—the DiDiR design does not require cross-sectional balance; it requires parallel trends in the post-treatment period, which is assessed in the event study (Figure 1).

Table B.13: Gelbach (2016) Decomposition: Channels of Price Effect

| | Coefficient | SE | % of gap |
|---|---|---|---|
| *Panel A: Overall effect* | | | |
| Short regression ($g65 \times Pre$) | -0.1318*** | (0.0096) | |
| Full regression ($g65 \times Pre$) | -0.1227*** | (0.0101) | |
| Gap (short − full) | -0.0091 | | 100.0% |
| *Panel B: Channel decomposition ($\delta_k = \gamma_k \times \pi_k$)* | | | |
| Competition (log firms) | 0.0078*** | (0.0010) | -85.0% |
| Composition (SME winner) | -0.0169*** | (0.0014) | 185.0% |
| Direct effect (unexplained) | -0.1227*** | (0.0101) | |
| Observations | | 648,616 | |
| Item FE | | YES | |

*Notes:* Gelbach (2016) decomposition of the price effect into channel contributions. Short regression includes only treatment and controls; full regression adds mediators. $\delta_k = \gamma_k \times \pi_k$ where $\gamma_k$ is the mediator coefficient in the full regression and $\pi_k$ is the treatment coefficient in the auxiliary regression of mediator $k$ on treatment. 18-month window, completed items. SE via delta method. *** $p$<0.01, ** $p$<0.05, * $p$<0.1.

Table B.14: Balance Test: Pre-Treatment Covariate Means by Group

| | Group 65 | | Other Groups | | | |
|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Difference | $p$-value |
| Log quantity | 5.08 | 2.89 | 3.10 | 2.27 | 1.985*** | 0.000 |
| Sealed bid (share) | 0.22 | 0.42 | 0.60 | 0.49 | -0.377*** | 0.000 |
| Completion rate | 0.67 | 0.47 | 0.77 | 0.42 | -0.101*** | 0.000 |
| Number of firms | 3.17 | 2.52 | 4.66 | 3.27 | -1.482*** | 0.000 |
| Number of valid bids | 11.10 | 15.49 | 10.13 | 15.74 | 0.970*** | 0.000 |
| Price (levels) | 1,402.09 | 20,563.54 | 1,465.85 | 50,385.93 | -63.755 | 0.622 |
| Distance (km) | 238.54 | 270.42 | 164.78 | 182.23 | 73.761*** | 0.000 |

*Notes:* Pre-treatment period means (September 2016 to February 2018) for Group 65 and all other groups. Difference = Group 65 − Others. $p$-values from Welch two-sample $t$-tests. Price and distance are conditional on item completion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## N. Excluding Last Post-Period Semester

Table B.16 re-estimates the main specifications using a truncated 18-month window that excludes the final post-period semester (March–August 2019), which showed a positive drift in the event study coefficients. The results are virtually unchanged from the full-window estimates, confirming that the main findings are not driven by the final semester's deviation from the parallel trends pattern.

Table B.15: Robustness: Excluding Last Post-Period Semester (Mar–Aug 2019)

| | Log prices | | Log firms | | Log bids | | Distance | |
|---|---|---|---|---|---|---|---|---|
| | Base | PBU FE | Base | PBU FE | Base | PBU FE | Base | PBU FE |
| $g65 \times Pre$ | -0.1300*** | -0.1297*** | 0.1031*** | 0.1091*** | 0.0512*** | 0.0608*** | 9.6580*** | 4.7026** |
| | (0.0116) | (0.0113) | (0.0063) | (0.0064) | (0.0082) | (0.0083) | (2.3462) | (2.3492) |
| Observations | 528,676 | 528,676 | 629,390 | 629,390 | 629,390 | 629,390 | 528,676 | 528,676 |
| $R^2$ | 0.9206 | 0.9256 | 0.5570 | 0.5955 | 0.5639 | 0.5945 | 0.3359 | 0.4267 |

*Notes:* Re-estimation of the 18-month window specification excluding the final post-period semester (March–August 2019). The effective window is September 2016 to February 2019. Standard errors clustered at the item level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## O. Score Cluster Bootstrap

Table B.17 reports inference from a score cluster bootstrap at the item-group level (76 clusters). Because the main specification clusters standard errors at the item level while treatment is assigned at the group level, this exercise verifies that the results survive inference appropriate for the number of treatment clusters. All four outcomes remain highly significant under the bootstrap distribution, with $p$-values below 0.001.

## P. Mechanism Evidence

Tables B.18–B.20 provide three pieces of mechanism evidence. Table B.17 splits the sample by item quantity at the median. The increase in participating firms is five times larger for high-quantity items (11.7% vs. 2.4%), consistent with the causal forest finding that item quantity is the dominant moderator of the treatment effect (variable importance

Table B.16: Score Cluster Bootstrap Inference (Group-Level, 76 Clusters)

| | Log prices | Log firms | Log bids | Distance |
|---|---|---|---|---|
| $g65 \times Pre$ | -0.1309*** | 0.0926*** | 0.0511*** | 10.8570*** |
| Group-clustered SE | (0.0178) | (0.0024) | (0.0028) | (0.4079) |
| Bootstrap $p$-value | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Bootstrap 95% CI | [-0.1487, -0.1131] | [0.0902, 0.0950] | [0.0482, 0.0539] | [10.4491, 11.2649] |
| Observations | 649,714 | 773,263 | 773,263 | 649,714 |

*Notes:* 18-month window, baseline specification (item FE). Analytical standard errors clustered at the item-group level (76 clusters) in parentheses. Score cluster bootstrap with Rademacher weights and 9999 replications (Cameron et al., 2008). $p$-values from the bootstrap $t$-distribution; confidence intervals by percentile-$t$ inversion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

= 0.486). The price effects are similar in magnitude across quantity levels, suggesting that even small increases in competition translate into meaningful price reductions.

Table B.18 implements a mediation analysis by sequentially adding the number of firms and the SME winner indicator as controls to the price regression. Controlling for both mediators attenuates the treatment coefficient by approximately 6%, indicating that most of the price effect operates through channels beyond simple participation counts and winner composition—such as bid aggressiveness conditional on entry or the efficiency of matched suppliers.

Table B.19 decomposes the group 65 effect into medications (class 6531, subject to CMED pharmaceutical price regulation) and other medical supplies (hospital equipment, dental supplies). The price effect is approximately 67% larger for medications ($-0.166$) than for other medical items ($-0.099$). This pattern could reflect either (a) greater market power in the oligopolistic pharmaceutical sector, making the SME restriction more costly, or (b) residual differential inflation from CMED-regulated prices. Both interpretations are consistent with the aggregate finding; the distinction bears on the precise mechanism. The firm participation effect is also substantially larger for medications (+0.152 vs. +0.044), supporting the competition channel.

Table B.17: Dose-Response: Effects by Item Quantity

| | Log prices | | Log firms | |
|---|---|---|---|---|
| | Base | PBU FE | Base | PBU FE |
| *Panel A: High quantity (above median)* | | | | |
| $g65 \times Pre$ | -0.1230*** | -0.1297*** | 0.1165*** | 0.1265*** |
| | (0.0111) | (0.0108) | (0.0070) | (0.0071) |
| Observations | 348,363 | 348,363 | 402,194 | 402,194 |
| *Panel B: Low quantity (below median)* | | | | |
| $g65 \times Pre$ | -0.1336*** | -0.1289*** | 0.0235*** | 0.0234*** |
| | (0.0117) | (0.0111) | (0.0077) | (0.0077) |
| Observations | 301,351 | 301,351 | 371,069 | 371,069 |

*Notes:* 18-month window. The sample is split at the median of log quantity (3.18). High-quantity items are more likely to attract large non-SME suppliers, so the SME restriction is more binding for these items. This split aligns with the causal forest finding that item quantity is the dominant source of treatment effect heterogeneity (variable importance = 0.486). SE clustered at item level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.18: Mechanism Decomposition: Entry Margin and Winner Composition

| | (1) Log price | (2) Log firms | (3) SME winner | (4) Price + firms | (5) Price + SME | (6) Price + both |
|---|---|---|---|---|---|---|
| $g65 \times Pre$ | -0.1309*** | 0.0926*** | -0.3858*** | -0.1395*** | -0.1159*** | -0.1227*** |
| | (0.0096) | (0.0059) | (0.0040) | (0.0099) | (0.0098) | (0.0101) |
| Log firms | | | | 0.1087*** | | 0.1099*** |
| | | | | (0.0106) | | (0.0106) |
| SME winner | | | | | 0.0390*** | 0.0438*** |
| | | | | | (0.0037) | (0.0037) |
| Observations | 649,714 | 773,263 | 649,714 | 648,616 | 649,714 | 648,616 |

*Notes:* 18-month window, item FE, SE clustered at item level. Columns (1)–(3): DiDiR for each outcome separately. Column (4): price regression adding log firms as control (mediation by competition). Column (5): price regression adding SME winner indicator (mediation by composition). Column (6): price regression with both mediators. The attenuation of the $g65 \times Pre$ coefficient from (1) to (4)–(6) measures the fraction of the price effect explained by each channel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.19: Within-Group-65 Heterogeneity: Medications vs. Other Medical Supplies

| | Log prices | | Log firms | |
|---|---|---|---|---|
| | Base | PBU FE | Base | PBU FE |
| *Panel A: Medications (class 6531)* | | | | |
| $g65_{med} \times Pre$ | -0.1658*** | -0.1744*** | 0.1516*** | 0.1605*** |
| | (0.0174) | (0.0172) | (0.0100) | (0.0101) |
| Observations | 573,080 | 573,080 | 678,547 | 678,547 |
| *Panel B: Other medical supplies (non-6531)* | | | | |
| $g65_{other} \times Pre$ | -0.0992*** | -0.0973*** | 0.0444*** | 0.0504*** |
| | (0.0092) | (0.0086) | (0.0057) | (0.0057) |
| Observations | 593,920 | 593,920 | 705,616 | 705,616 |

*Notes:* 18-month window. Panel A restricts group 65 to class 6531 (medications, subject to CMED pharmaceutical price regulation), using all non-group-65 items as the comparison. Panel B restricts group 65 to non-medication medical supplies (hospital equipment, dental supplies, etc.). If the price effect were driven by differential pharmaceutical inflation rather than competition, it should appear primarily in Panel A. SE clustered at item level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## Q. Synthetic Control

As an alternative to the DiDiR design, I estimate a synthetic control model (Abadie et al., 2010) using the 68 balanced product groups as donors to construct a synthetic group 65. The augmented synthetic control (Ben-Michael et al., 2021) uses Ridge regularization to improve pre-treatment fit. The treatment date is March 2018, when group 65 lost its open-tender exemption.

Figure A.15 shows that the synthetic control matches group 65 prices almost exactly in the pre-treatment period (L2 imbalance $\approx 0$). After March 2018, group 65 prices rise relative to the synthetic by approximately 17 log points on average (Table B.20), consistent in direction and magnitude with the DiDiR estimates of 13 log points. The post-treatment divergence reflects the cost of losing the open-tender exemption: once group 65 became subject to SME-only tender rules, its prices converged upward toward the always-treated groups.

Figure A.16 plots the gap over time. The in-space placebo test (Figure A.17) assigns placebo treatment status to each of the 68 donor groups: only 7 of 68 placebos produce an ATT as large as group 65's (rank $p$-value = 0.10), providing moderate evidence that the estimated effect is not a statistical artifact.

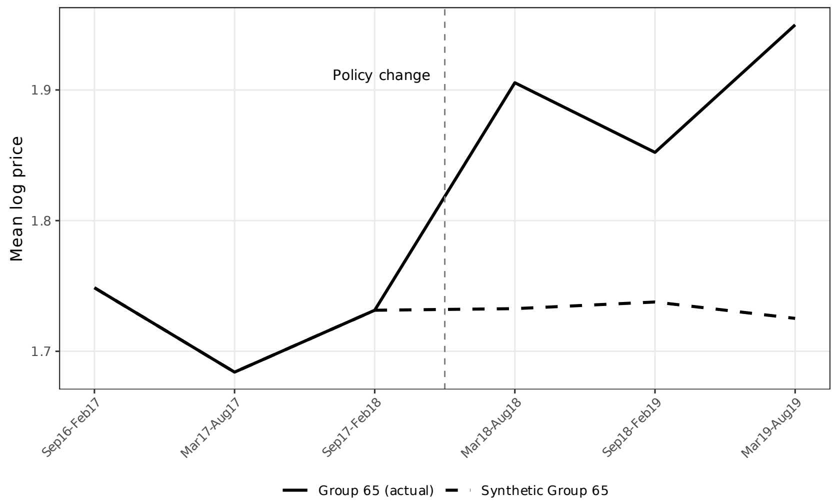


Figure A.15: Synthetic Control: Group 65 (Actual) vs. Synthetic Group 65

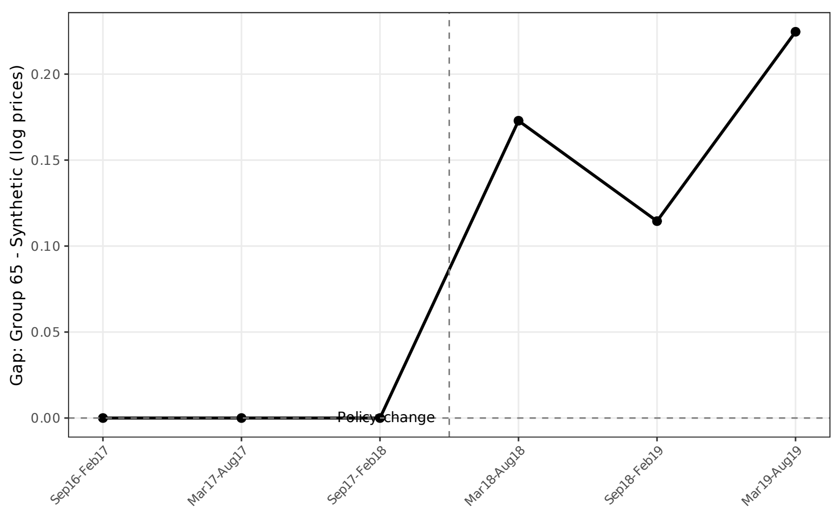


Figure A.16: Synthetic Control: Gap Between Group 65 and Synthetic (Log Prices)

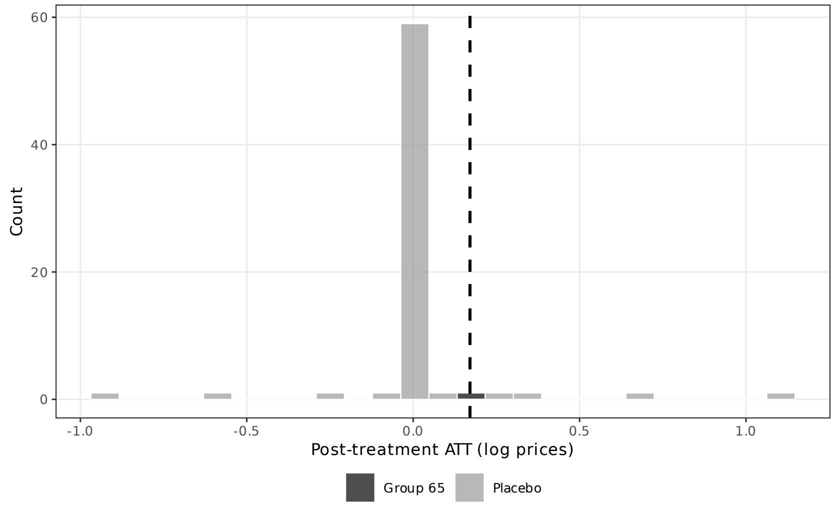


Figure A.17: Synthetic Control: In-Space Placebo Distribution of Post-Treatment ATTs

Table B.20: Synthetic Control: Summary Results

| | |
|---|---|
| *Panel A: Gaps* | |
| Mean pre-treatment gap (semesters 1–3) | 0.0000 |
| Mean post-treatment gap (semesters 4–6) | 0.1707 |
| Post-treatment ATT (mean) | 0.1707 |
| Placebo rank $p$-value | 0.103 (7/68) |
| L2 imbalance | 0.0000 |
| *Panel B: Top donor weights (> 1%)* | |
| Product group | Weight |
| Group 40 | 0.044 |
| Group 75 | 0.099 |
| Group 79 | 0.382 |
| Group 89 | 0.436 |
| Other groups | 0.039 |
| Balanced groups | 69 |
| Semesters | 6 |

*Notes:* Augmented synthetic control with Ridge augmentation. Outcome: mean log price at the group × semester level. Treatment date: March 2018 (group 65 loses open-tender exemption). Pre-treatment gap measures how much lower group 65 prices were under open tenders relative to the synthetic counterfactual. Post-treatment gap near zero validates the parallel trends assumption. Inference via conformal method.